

ME 204 A : MECHANICS OF SOLIDS

2L, 1T 3 Hours, 100 Marks

Direct Stresses in Machine Parts : Analysis of stress and strain. -Hooke's law, stress-strain diagram, Poisson's ratio, Modulus of elasticity, Relation between elastic constants

Simple stresses : Tensile, compressive, shear, crushing and bearings. criteria. Hoop stress in thin cylindrical and spherical vessels subjected to internal pressure

Machine Members in Bending : Theory of bending of straight and curved beams. Bending and shear stress distribution in beams. Shear force and bending moment diagrams. Moment of inertia of different machine element sections

Deflection and stiffness : Mecauly's method, area moment method, numerical integration method and energy method Numerical problems of line shafts under simple loading with pulleys, ears, flywheel etc. Brief introduction to analysis of shafts with three Supports.

Machine Members in Torsion : Introduction to torsion of circular solid and hollow sections. Angular deformation and strain energy.

Stress in Slender Machines Members under axial Comressive Loads : Analysis of long and short columns. Euler's theory and effect of end conditions on stresses. Emperical formulas

Machine Members subjected to Combined Loading: Combination of axial, bending, and torsional stresses. Concept of principal stresses *and* planes. Mohr's circle of stresses. 2D state of strain, Plane stress and Plane strain conditions. Mohr's circle for strain. Theories of failure. Simple problems of line shaft.

SN Name of Authors /Books /Publisher

Year of

Pub.

1 Timoshenko, S.P., and Gere, J.M., “Mechanics of Materials”, 2nd Ed., CBS Publishers

2 Crandall, S.H., Dahl, N.C., and Lardner, T.J., “An Introduction to the Mechanics of Solids”, TMGH

3 Strength of materials , Ramamurtham

4 Punmia, Jain and Jain, “Mechanics of Materials”, Laxmi Publication

5 Popov, E.P., Nagarajan, S., and Lu, Z. A., “Mechanics of Materials”, Prentice-Hall of India

INTRODUCTION

In all engineering construction, the component parts of a structure or a machine must be assigned definite physical sizes. Such parts must be properly proportioned to resist the actual or probable forces that may be imposed upon them.

- Thus for example the walls of a pressure vessel (Such as boiler) must be of adequate strength to withstand the internal pressure.
- The floors of a building must be sufficiently strong for their intended purpose.
- The shaft of a Machine must be of adequate size to carry the required torque.
- A wing of an airplane must safely withstand the aerodynamic loads that may come upon it during take off, flight and landing.
- Likewise the parts of a composite structure must be rigid enough so as not to deflect excessively when in operation under the imposed loads.

The ability to determine the minimum load that an element of a structure can carry before failure happens is of great importance. In engineering practice such requirements must be met with the minimum expenditure of a given material.

- The subject strength of materials' involves analytical methods for determining the strength, stiffness and stability of various load carrying materials.
- This subject is very important to be studied by many braches of engineering. It is important for *civil engineers* in the design of bridges and buildings, for *nuclear engineers* in the design of reactor components, for *mechanical engineers* for the design of machineries and pressure vessels, for *metallurgical engineers*, who need fundamental concepts of this subject in order to understand how to improve existing materials further.

1. Simple Stresses and Strains

1.1 Introduction

Whenever a load is applied to a body, its shape changes i.e. the body gets deformed and the process is called deformation. The amount by which body is deformed depends upon the amount of load and the nature as well as the geometrical properties. It has been experimentally found that the cohesive forces between molecules of a body resist the deformation and force of resistance increases with increment in deformation. It has also been observed that the process of deformation stops when the force of resistance is equal to the external load. If the force of resistance is less than the external load, the material gets failed.

1.2 Stress: When an external force or load is applied on a body internal resistance is set up. Let us consider a bar acted upon by a Force P along its axis as in Fig 1.1(a). If the bar AC is cut in two sections at B, it results in Fig 1.1(b). The externally applied forces to one side of the cut part must be balanced by internal forces developed at the cut.

If the internal resistance offered by the section against the deformation be assumed to be uniform across the section, the intensity of resistance per unit of area is called the intensity of stress or unit stress. In general usage the word stress is used to mean the intensity of stress.

$$\text{Intensity of stress} = \sigma = \frac{F}{A} \quad \dots\dots(1.1)$$

$$\sigma = \text{Stress N/m}^2$$

$$A = \text{Cross Section area in m}^2$$

$$F = \text{Total force acting through the centroidal axis of the body in N.}$$

1.3 Strain: Strain is defined as the ratio of deformation of a body due to external load to the original length. It is a dimensionless parameter. If a load P is applied to a body the body deforms by δ and the original length be L the strain (ϵ) is given by

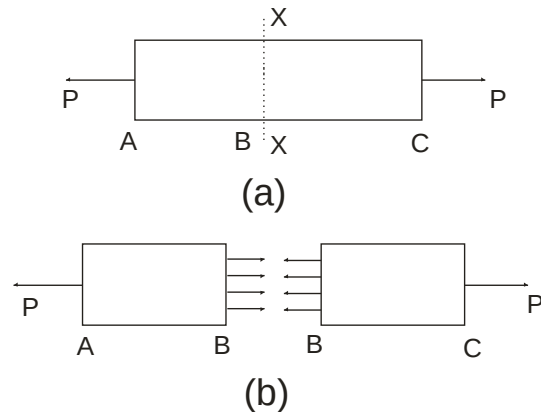


Fig. 1.1

$$\epsilon = \frac{\delta}{L} \quad \dots(1.2)$$

1.4 Simple Stress & Strain:

(a) Tensile Stress (σ_t): When a body is subjected to two equal and opposite pulls and the length of the body increases, the stress induced is called tensile stress (Fig 1.2).

The stress in (σ_t) simple tension in a body is equal to the external force P divided by the resisting area and in this case the resisting area is cross-sectional area A of the member in m^2 . Thus

$$\sigma_t = \frac{P}{A} \text{ N/m}^2 \quad \dots(1.3)$$



Fig. 1.2

Corresponding strain is called tensile strain

Tensile strain = $\frac{\text{Increase in length}}{\text{Original length}}$

$$e = \frac{dl}{l} \quad \dots(1.4)$$

(b) Compressive Stress (σ_c): When a body is subjected to two equal and opposite pushes and length of the body reduces, the stress induced is called compressive stress.

(Fig 1.3). For a straight short compression member in which the resultant load P acts through centre of gravity of every cross section the stress σ_c



Fig. 1.3

$$\sigma_c = \frac{P}{A} \quad \dots(1.5)$$

Corresponding strain is called compressive strain.

Compressive strain = $\frac{\text{Decrease in length}}{\text{Original length}}$

$$e = \frac{dl}{l} \quad \dots(1.6)$$

(C) Shear Stress (τ): When equal and opposite forces are applied tangentially to any cross sectional plane of a body, tending to slide its one part over other, the body is said to be in state of shear. The corresponding stress induced is shear stress. In Fig.1.4 riveted

joint tends to shear off at section XX due to externally applied load P. Shear stress is produced at the cross section of rivet. If the cross section area of rivet be A shear stress

$$\tau = \frac{P}{A} \quad \dots(1.7)$$

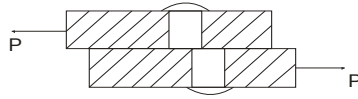


Fig. 1.4

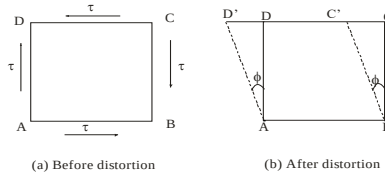


Fig. 1.5

Shear Strain: It is a measure of the angle through which a body is deformed by the applied load. Let a rectangular block ABCD, fixed at bottom is distorted through an angle ϕ by the externally applied tangential load P to face CD as in Fig. 1.5. The face CD moves to new position C'D'. Let the movement of end C is very small.

$$\text{Then shear strain} = \frac{CC'}{CB} = \tan \phi = \phi \text{ radian} \quad [\because \phi \text{ small}]$$

1.5 Elasticity: A material is composed of small particles and molecules, between which forces are acting. These molecular forces resist the change in the shape of the body which external forces tend to produce. Under the action of external forces the particles of body are displaced and the displacement continues till the equilibrium is established between the external and internal forces.

As in Fig. 1.6 a bar is loaded at the end. Under the action of this load a certain elongation of the bar will take place. When the load is diminished the elongation of the bar diminishes and the loaded end of the bar moves upward. The property by which a body returns to its original shape after removal of the load is called **elasticity**. The body is *perfectly elastic* if it recovers its original shape completely after unloading, *partially elastic* if the deformation disappear partially after unloading and *plastic* if the deformation is permanent.

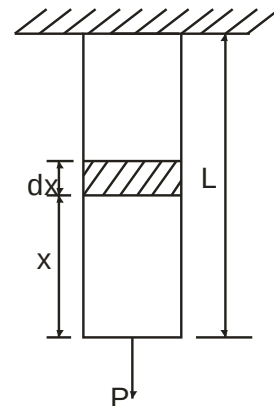


Fig. 1.6

1.6 Hooke's Law

It states that within an elastic limit stress is proportional to strain.

$$\begin{array}{l} \text{Stress} \propto \text{Strain} \\ \text{Stress} \\ \text{or} \quad \frac{\text{Stress}}{\text{Strain}} = E \quad [C \text{ is a constant}] \end{array}$$

‘E’ is known as coefficient of elasticity or modulus of elasticity. The ratio of tensile stress and tensile strain or compressive stress and compressive strain is called young’s modulus of elasticity and is denoted by-.

$$\frac{\sigma_t}{\epsilon_t} \quad \text{or} \quad \frac{\sigma_c}{\epsilon_c} = E \quad \dots(1.8)$$

The ratio of shear stress and shear strain is called shear modulus or modulus of rigidity. It is denoted by C or G_t

$$\frac{\tau}{\epsilon_s} = C \quad \dots(1.9)$$

Let us consider a body subjected to a tensile force P, having length l and cross section area A, the stress induced

$$\sigma = \frac{P}{A}$$

And if the δ is the deformation and strain

$$\epsilon_t = \frac{\delta}{l}$$

As per Hooke’s law $\frac{\sigma}{\epsilon_t} = E$

$$\frac{Pl}{A\delta} = E$$

$$\begin{array}{l} \text{or deformation} \quad \delta = \frac{Pl}{AE} \quad \dots(1.10) \end{array}$$

1.7 Stress Strain Curve For Mild Steel: Curve (Fig. 1.7) for a round mild steel bar of which ordinate represents the stress intensities and the abscissa the corresponding strains.

The points worth noting are:

- ❖ The line OA being straight. OA is called the proportional limit upto which Hooke's law is applicable i.e. stress is proportional to strain up to A.
- ❖ Mild steel remains elastic a little beyond A i.e. upto point B, so the proportional limit and elastic limit are not same for mild steel but same for rest of the materials.
- ❖ Upon loading beyond this point the elongation increases more rapidly and the diagram becomes curved. At B a sudden elongation of the bar takes place without an appreciable increase in the tensile force. The phenomenon is called 'Yielding' of the metal. The stress corresponding to this 'B' point is called yield point stress.

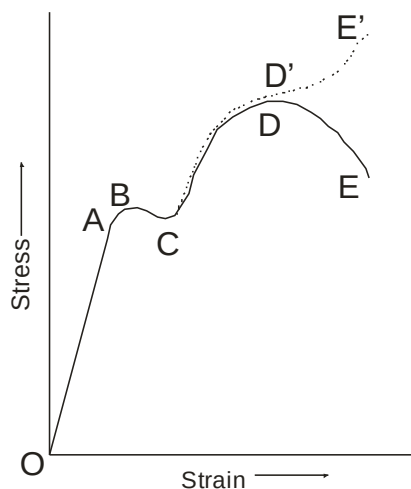


Fig. 1.7

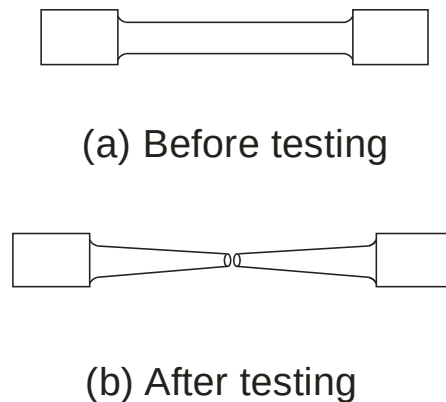


Fig. 1.8

- ❖ Upon further stretching of the bar, the material recovers its resistance and the tensile force increases with the elongation upto point 'D', where the force attains its maximum value and the process of necking begins.

The corresponding stress is called the 'Ultimate Stress'. Necking is displayed by contraction of the specimen. Fig. 1.8.

- ❖ Beyond the point D elongation of the bar takes place with a diminution of the load and fracture finally occurs at a load corresponding to point E of diagram. The corresponding stress is called the 'Breaking stress'.

The stress-strain curve is drawn experimentally and it is unique for every metal. Hence the points observed on the curve are the properties of the metal and unique for every metal. So the point A corresponds to **limit of elasticity**, point B corresponds to **Yield strength** and point D correspond to **Ultimate strength** of the metal.

Stretching of this bar is accompanied by lateral contraction but it is established for calculating the yield & ultimate point stress, hence initial cross-sectional area is used. If the load is divided by the original area of cross section it results in nominal stress which is less at breaking load than at the maximum load as indicated by E&D respectively. If actual area of cross section is used to find stress intensity, rupture or breaking stress is greater than at the maximum load. The dashed portion of the curve shows relation between actual stress intensity and strain. This is called true stress strain curve.

Fig. 1.9 represents a tensile test diagram for cast iron. The material has a very low proportional limit and has no definite yield point. For brittle materials stress-strain curve is as in Fig. 1.9. There is no yield point in case of brittle material.

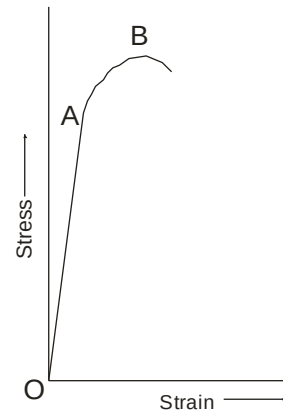


Fig. 1.9

1.8 Factor Of Safety And Working Stresses:

Working stress: The greatest calculated stress to which a part of machine or structure is ever subjected is called the working stress or allowable stress or permissible stress or design stress and the ratio

$$\frac{\text{Ultimate Strength}}{\text{Working Stress}} \text{ or } \frac{\text{Yield Stress}}{\text{Working Stress}} \quad \dots(1.11)$$

is called Factor of Safety. The ultimate or yield strength is the property of the metal and it is known while selecting the metal for design purpose. For design calculations the factor

of safety is to be assumed and design stress is to be calculated. Hence from the equation

$$(1.11) \text{ working stress is the ratio } \frac{\text{Ultimate Strength}}{\text{Factor of safety}} \text{ or } \frac{\text{Yield Stress}}{\text{Factor of safety}} \dots(1.12)$$

Factor of safety depends upon:

1. **Nature of loading:** The factor of safety varies very greatly according to the nature of the stresses, whether constant, variable or alternating simple or compound.
2. **Homogeneity of materials used:** The factor of safety will be higher if the quality of material used is not of uniform properties.
3. **Degree of safety required:** If safety required is more than factor of safety will be higher. For example the areas where human hazards are involved we would require more safety like boiler applications.
4. **Degree of economy desired:** If we go for higher factor of safety the cost of the component will be more. Hence economical constraints are equally important while considering factor of safety.
5. **The accuracy of problem formulations:** If the estimation of stresses and load is not accurate we have to take higher factor of safety.
6. **Manufacturing processes:** If the quality of manufacturing methods used is good factor of safety required is lesser.

1.9 Elongation of Bars of Varying Cross Section

Fig 1.10 Shows a bar consisting of three lengths l_1 , l_2 , l_3 with cross section areas A_1 , A_2 , and A_3 , Subjected to an axial load P . Though the force on each section is same stress induced in each section is different.

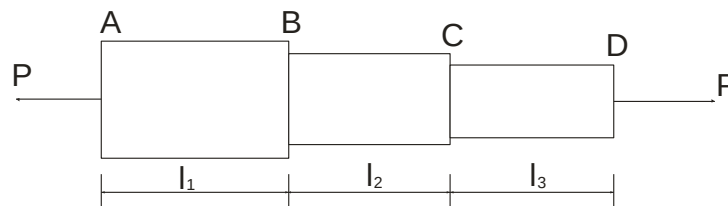


Fig. 1.10

Intensity of stress for section AB = $\sigma_1 = P/A_1$

Similarly intensity of stress for section BC = $\sigma_2 = P/A_2$
 and intensity of stress for section CD = $\sigma_3 = P/A_3$

$$\therefore \text{Strain of section AB} = \epsilon_1 = \frac{\sigma_1}{E}$$

Where E – Modulus of elasticity

$$\therefore \text{Strain of section BC} = \epsilon_2 = \frac{\sigma_2}{E}$$

$$\therefore \text{Strain of section CD} = \epsilon_3 = \frac{\sigma_3}{E}$$

$$\text{Change in length of section AB} = \delta l_1 = \epsilon_1 l_1 = \frac{Pl_1}{A_1 E}$$

$$\text{Change in length of section BC} = \delta l_2 = \epsilon_2 l_2 = \frac{Pl_2}{A_2 E}$$

$$\text{Change in length of section CD} = \delta l_3 = \epsilon_3 l_3 = \frac{Pl_3}{A_3 E}$$

$$\begin{aligned} \therefore \text{Total change in length of bar} &= \frac{\delta l_1 + \delta l_2 + \delta l_3}{P} \left[\frac{l_1}{A_1} + \frac{l_2}{A_2} + \frac{l_3}{A_3} \right] \\ &= \frac{P}{E} \left[\frac{l_1}{A_1} + \frac{l_2}{A_2} + \frac{l_3}{A_3} \right] \end{aligned} \quad \dots(1.13)$$

1.12 Principal Of Superposition

Concept of free body diagram

If an elastic body is subjected to a no. of direct stresses at different sections along the length of the body, the deformation of individual sections can be very easily found if free body diagrams are drawn for the individual sections. According to principle of superposition the total deformation of the body will then be equal to the algebraic sum of the deformations of the individual sections. For example let us consider a bar subjected to different forces at different sections as in Fig. 1.14 (a). To draw the free body diagrams for AC all the forces to the left of the normal sections A to be considered. The net force is 20N, tensile considering the net force to right of C is $18 + 6 - 4 = 20\text{N}$.

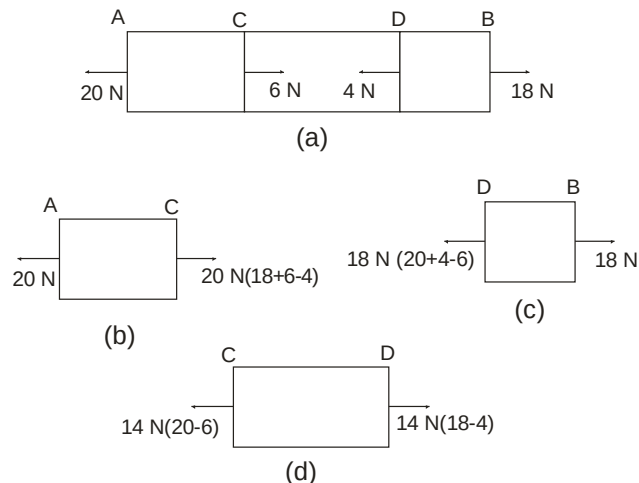


Fig. 1.14

Thus the section AC of length l_1 , is subjected to tensile force of 20 N, as in Fig. 1.14 (b).

Similarly, for the section CD of length l_2 consider the algebraic sum of the forces to the left of normal section at C or the sum of the forces to the right of the normal section at D.

Net force on CD = $20 - 6 = 14$ or $18 - 4 = 14$. (Fig. 1.14 (d))

Similarly section DB is subjected to 18 N (Fig. 1.14 (c))

The extension of AC $\Delta_1 = \frac{P l_1}{AE} = \frac{20 l_1}{A_1 E}$

$$\text{CD} \quad \Delta_2 = \frac{14 l_2}{A_2 E}$$

$$\text{DB} \quad \Delta_3 = \frac{18 l_3}{A_3 E}$$

Total extension of the bar AB will be given by

$$\Delta = \Delta_1 + \Delta_2 + \Delta_3 \quad \dots\dots(1.17)$$

3 Shear Force and Bending Moment

3.1 Introduction

A beam is a structural member subjected to a system of external forces at right angles to its axis. The process of deformation of a bar because of the forces transverse to its axis is called bending. The algebraic sum of the vertical forces at any section of a beam to the right or left of the section is known as shear force. It is briefly written as S.F. The algebraic sum of the moments of all the forces acting to the right or left of the section is known as bending moment. It is written as B.M. idered.

3.2 Shear Force And Bending Moment Diagrams

A shear force diagram is one which shows the variation of the shear force along the length of the beam. A bending moment diagram is one which shows the variation of the bending moment along the length of the beam.

3.3 Types Of Loads

A beam is normally horizontal and the loads acting on the beams are generally vertical. The following are the important types of load acting on a beam:

3.3.1. Concentrated Or Point Load. A concentrated load is one which is considered to act at a point. Although in practice it must really be distributed over a small area because such knife edge contacts are neither possible nor desirable. In Fig. 3.1(a) W shows the point load acting on a beam.

3.3.2. Uniformly Distributed Load. A uniformly distributed load is one which is spread over a beam in such a manner that intensity of loading w is uniform along the length (*i.e.* each unit length is loaded to the same rate) as shown in Fig. 3.1(b). The intensity of loading is expressed as w N/m run.

3.3.3. Uniformly Varying Load. A uniformly varying load is one which is spread over a beam in such a manner that intensity loading varies from point to point along the beam as shown

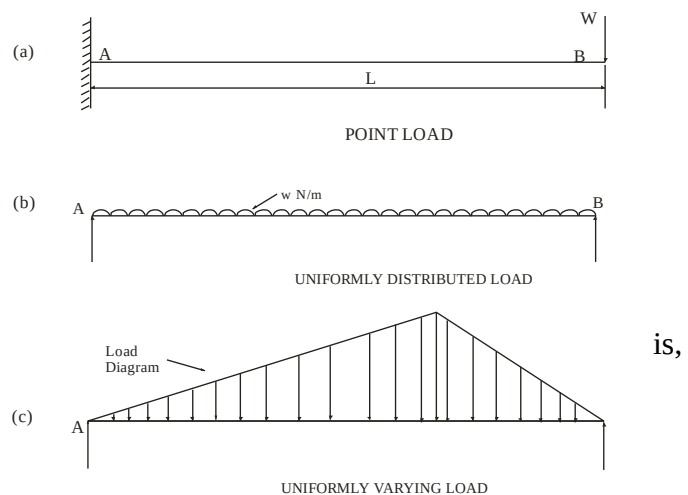


Fig. 3.1

in Fig. 3.1(c) in which load is zero at one end and increases uniformly to one point and then decreases to zero to the other end.

3.4 Types Of Beams

The following are the important types of beams:

3.4.1. Cantilever Beam. A beam which is fixed at one end and free at other end is known as cantilever beam. Such beam is shown in Fig. 3.2(a).

3.4.2. Simply Supported Beam. A beam supported or resting on the supports at its both ends is known as simply supported beam. beam is shown in Fig. 3.2(b).

3.4.3 Overhanging Beam. If end portion of a beam is ex-tended beyond the support, such beam is known as overhanging beam. Overhanging beam is shown in Fig. 3.2(c).

3.4.4. Fixed Beams. A beam, whose both ends are fixed or built in walls, is known as fixed beam. Such beam is shown in Fig. 3.2(d). A fixed beam is also known as a *built-in*, or encastred beam.

3.4.5. Continuous Beam. A beam which is provided more than two supports or more than one span as shown in Fig. 3.2(e) is known as continuous beam.

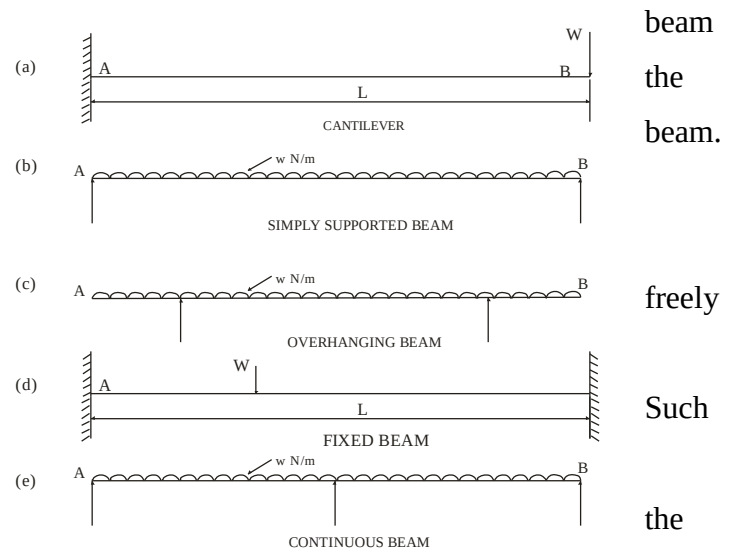


Fig. 3.2

3.5 Shear Force and Bending Moment

Fig. 3.3(a) shows a simply supported of span L. A single point load Pacts at a distance of L_1 from left support. The two reactions at left and right supports may be calculated from equations of static equilibrium. These equations state that the algebraic sum of all forces acting along two mutually perpendicular directions is zero. In case of beams one direction is taken along the beam axis and second direction is transverse to the beam axis. The equations of equilibrium further state that algebraic sum of moments of all forces acting on a body taken about any point in the plane of forces will vanish.

Taking moment about right support:

$$R_1 L = W (L - L_1) \text{ or } R_1 = \frac{W(L - L_1)}{L}$$

Summing up the forces transverse to the beam axis

$$R_2 - W + R_1 = 0 \text{ or } R_1 = \frac{WL_1}{L}$$

Where R_1 & R_2 are the support reactions.

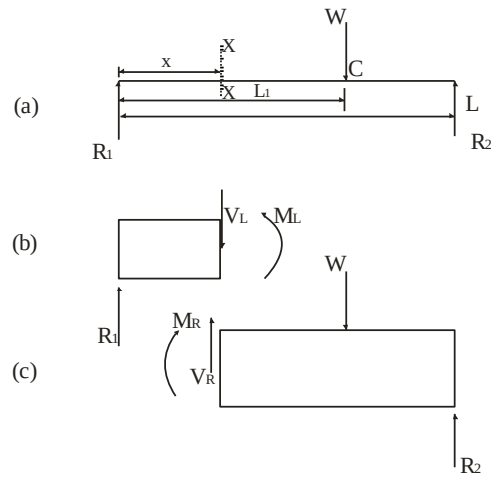


Fig. 3.3 of

Fig. 3.3(b) shows the free body diagram left hand portion of the beam cut by a transverse plane XX at a distance of x from left support. It can be seen that to balance R_1 on the right side of the cut portion a downward internal force V_L equal and opposite to R_1 ($V_L = -R_1$) will act on the section at a distance x from the left support. The force V_L is called shear force. R_1 and V_L will form a clockwise couple $R_1 x$ and hence an internally developed moment i.e. moment of resistance ($M_L = -R_1 x$) will have to balance the moment $R_1 x$ and will act anticlockwise. (We are taking clockwise moment positive.)

Thus it now becomes clear that in a loaded beam at any section the static equilibrium is achieved by two internal reactions viz a shear force V_L and a moment of resistance M_L . By drawing a free body diagram of right hand portion of the beam as in Fig. 3.3(c) it can be shown that to balance $(-W + R_2)$ on the left side of the cut portion a upward internal force V_R equal and opposite to $(-W + R_2)$ will act on the section.

$$V_R = -(-W + R_2) = +R_1$$

The force V_R is called shear force. $(-W + R_2)$ and V_R will form a anticlockwise couple:

$$M = W (L_1 - x) - \frac{WL_1}{L} (L - x) = -\frac{WL_2}{L} x = -R_1 x$$

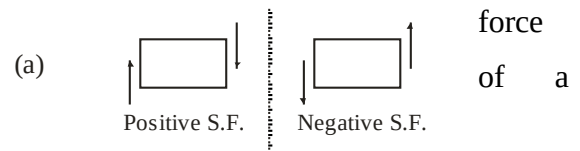
Hence an internally developed moment i.e. moment of resistance ($M_R = R_1 x$) will have to balance the above moment $R_1 x$ and will act clockwise. (We are taking clockwise moment positive). We can observe that V_L and V_R are same in magnitude but opposite in signs and acting

on opposite sides of the sectioning plane. Similarly M_L and M_R are same in magnitude but opposite in directions.

‘The **shear force** at a section of a loaded beam is the algebraic sum of all the forces acting either left or right side of the section. Similarly **bending moment** at a section is the algebraic sum of moments of all the forces either to the right or to the left of the section taken about the section.’

3.6 Sign Conventions:

(a) **Shear Force:** shear force having a downward direction to the right hand side of a section will be taken as positive. Similarly shear force having an upward direction to the right hand side of a section will be taken as negative. (Fig. 3.4(a))



(b) **Bending Moment:** A bending moment causing convexity upwards will be taken as negative and will be called a hogging. Similarly a bending moment causing concavity upwards will be taken as positive and will be called a sagging. (Fig. 3.4(b))

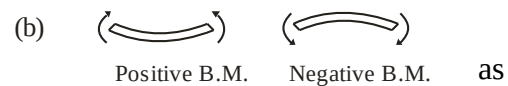


Fig. 3.4

3.7 Important Points For Drawing Shear Force And Bending Moment Diagrams

The shear force diagram is one which shows the variation of the shear force along the length of the beam. A bending moment diagram is one which shows the variation of the bending moment along the length of beam. In these diagrams, the shear force or bending moment are represented by ordinates whereas the length of the beam represents abscissa.

The following are the important points for drawing shear force and bending moment diagrams:

1. Consider the left or the right portion of the section.
2. Add the forces (including reaction) normal to the beam on one of the portion. If right portion of the section is chosen, a force on the right portion acting downwards is positive while a force on the right portion acting upwards is negative.

If the left portion of the section is chosen, a force on the left portion acting upwards is positive while a force on the left portion acting downwards is negative.

3. The positive values of shear force and bending moments are plotted above the base line, and negative values below the base line.
4. The shear force diagram will increase or decrease suddenly i.e. by a vertical straight line at a section where there is a vertical point load.
5. The shear force between any two vertical loads will be constant and hence the shear force diagram between two vertical loads will be horizontal.
6. The bending moment at the two supports of a simply supported beam and at the free end of a cantilever will be zero.

3.8 Shear Force And Bending Moment Diagrams For A Cantilever

3.8.1 A Cantilever With A Point Load At The Free End

Fig. 3.5 (a) shows a cantilever AB of length L fixed at A and free at B and carrying a point load W at the free end B. Let us take a section X at a distance x from the free end.

Let F_x = Shear force at X, and

M_x = Bending moment at X.

S.F.D: Considering the right portion of the section the shear force at this section is equal to the resultant force acting on the right portion at the given section. The resultant force acting on the right portion at the section X is W which is acting in the downward direction. Since a force on the right portion acting downwards is considered positive, hence shear force at X is positive.

$$\therefore F_x = +W$$

The shear force will be constant at all sections of the cantilever between A and B as there is no other load between A and B. The shear force diagram is shown in Fig. 3.5 (b).

B.M.D.

The bending moment section X is given by

$$M_x = -W \times x \quad \dots(i)$$

(Bending moment will be negative as B.M. is of hogging nature, bending the

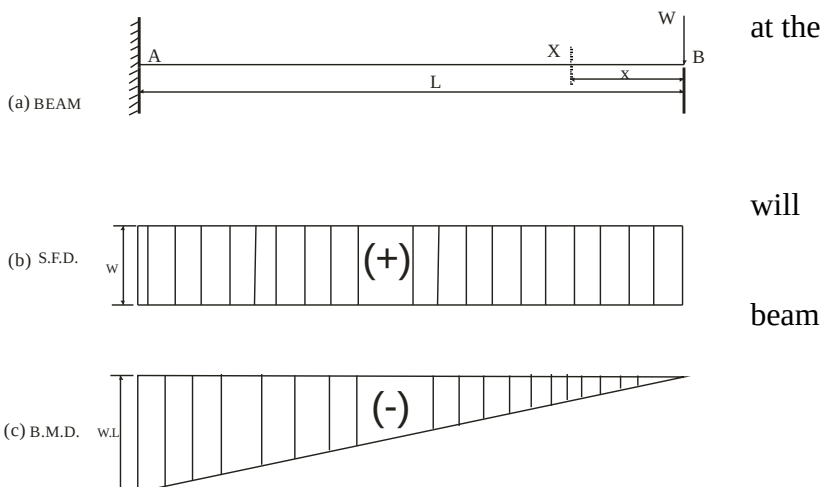


Fig. 3.5

convex upwards.)

From equation (i), it is clear that B.M. at any section is proportional to the distance of the section from the free end.

At $x=0$ i.e. at B, B.M. = 0

At $x=L$ i.e. at A, B.M. = $W \times L$

Hence B.M. follows the straight line law. The B.M. diagram is shown in Fig. 3.5 (c).

3.8.2 A Cantilever With Several Point Loads

S.F.D.: Between B and D, $F_x = +4$ (constant)

Between D and C, $F_x = +4 + 4 = 8$ (constant)

Between C and A, $F_x = +4 + 4 + 2 = 10$ (constant)

Therefore the S.F.D. consists of rectangles of different sizes as in Fig. 3.6 (b).

B.M.D.: Between B and D, $M_x = -4x$

At B, $x = 0 \therefore M_B = 0$.

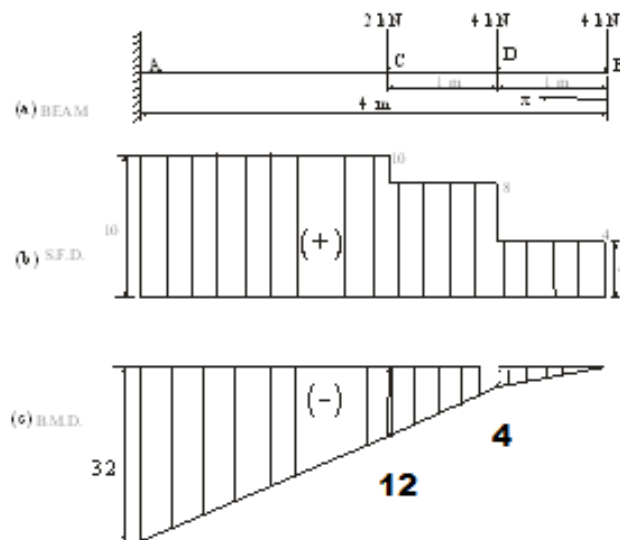


Fig. 3.6

At D, $x = 1 \text{ m} \therefore M_D = -4 \times 1 = -4 \text{ kN-m}$.

Between D and C, $M_x = -4x - 4(x - 1)$

At D, $x = 1 \text{ m} \therefore M_D = -4 \text{ kN-m}$.

At C, $x = 2 \text{ m} \therefore M_C = -12 \text{ kN-m}$.

Between C and A, $M_x = -4x - 4(x - 1) - 2(x - 2)$

At C, $x = 2 \text{ m} \therefore M_C = -12 \text{ kN-m}$.

At A, $x = 4 \text{ m} \therefore M_A = -32 \text{ kN-m}$.

Therefore the B.M.D. consists of lines of various slopes as in Fig. 3.6 (c).

3.8.3 A Cantilever With A Uniformly Distributed Load

Fig. 3.7 shows a cantilever of length L fixed at A and carrying a uniformly distributed load of w per unit length over the entire length of the cantilever.

S.F.D: The shear force at any section X at a distance x from the free end B will be equal to the resultant force acting on the right portion of the section. This resultant force is acting downwards. Resultant force on the right portion acting downwards is considered positive. Hence shear force at X is positive.

The resultant force on the right portion = $w \times \text{Length of right portion} = +w.x$.

At B, $x = 0 \therefore F_B = 0$.

At A, $x = L \therefore F_A = +wL$.

Therefore the S.F.D. consists of triangle as shown in Fig. 3.7(b).

B.M.D.:

To find the effect of the uniformly distributed load over a it is converted into point load at the C.G. of the uniformly distributed load.

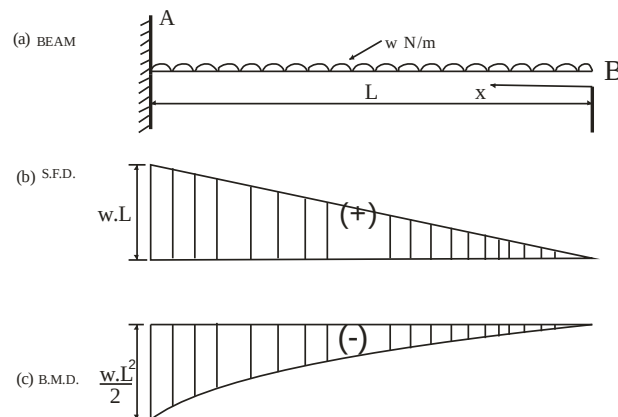


Fig. 3.7

section
acting

The bending moment at the section X is given by

$M_x = -(\text{Total load on right portion}) \times \text{Distance of section from the C.G. udl}$

$$= - (w \cdot x) \cdot \frac{x}{2}$$

$$= - w \cdot x \cdot \frac{x}{2} = - w \cdot \frac{x^2}{2}$$

.... (i)

(The bending moment will be negative as for the right portion of the section, the moment of the load at x is clockwise. Also the bending of cantilever will take place in such a manner that convexity will be at the top of the cantilever).

From equation (i), it is clear that B.M. at any section follows a parabolic law.

At B, $x = 0$ hence $M_B = 0$

At A, $x = L$ hence $M_A = -w \cdot \frac{L^2}{2}$

The bending moment diagram is drawn as shown in Fig. 3.7(c).

3.8.4 A Cantilever With A U.D.L. On Part Span From The Free End

S.F.D:

For section BC, $F_x = +w \cdot x$

At B, $x = 0 \therefore F_B = 0$.

At A, $x = a \therefore F_C = +w \cdot a$.

For section CA, $F_x = +w \cdot a$

(Constant)

At A, $x = L \therefore F_A = F_C =$

The S.F.D. is triangular from B to C and rectangular from C to A as shown in Fig. 3.8 (b).

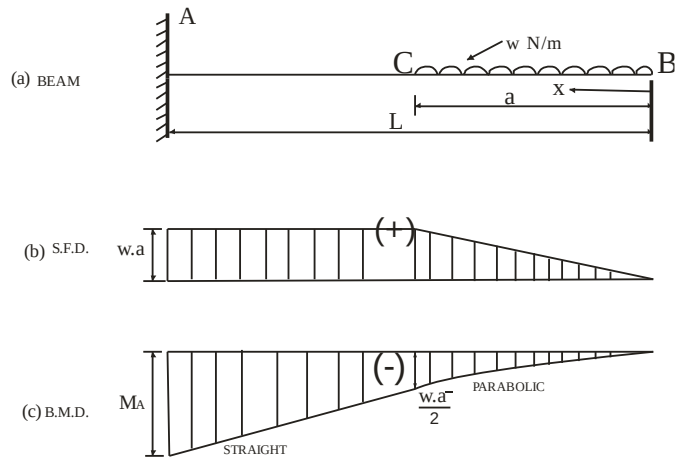


Fig. 3.8

+w.a.
B to C
as

B.M.D.: For section BC, $M_x = -w \cdot \frac{x^2}{2}$ (Parabolic)

At B, $x = 0 \therefore M_B = 0$.

At C, $x = a \therefore M_C = -w \cdot \frac{a^2}{2}$.

For section CA, $M_x = -w \cdot a \cdot \left(x - \frac{a}{2}\right)$ (linear)

At C, $x = a \therefore M_C = -w \cdot \frac{a^2}{2}$.

$$\text{At A, } x = L \therefore M_A = -w.a \left(L - \frac{a}{2} \right).$$

The B.M.D. is parabolic from B to C and linear from C to A as shown in Fig. 3.8 (c).

3.8.5 A Cantilever With Combination of Loads (U.D.L. and Point loads)

S.F.D:

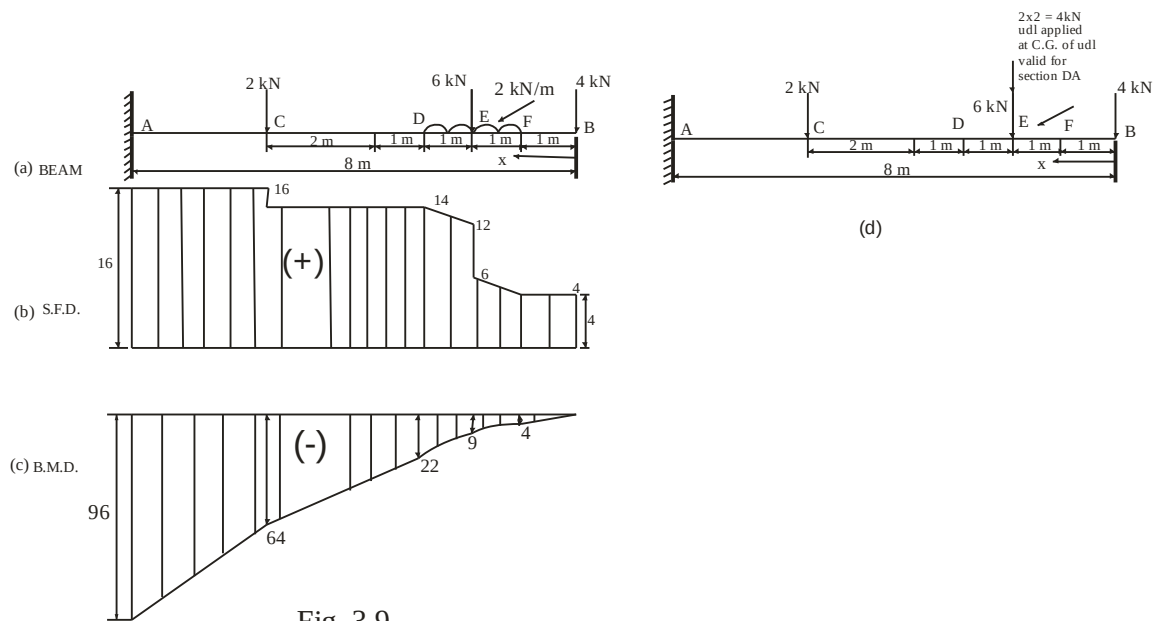


Fig. 3.9

For section BF, $F_x = +4$ kN (Constant)

At B, $x = 0 \therefore F_B = +4$ kN

At F, $x = 1 \therefore F_F = +4$ kN

For section FE, $F_x = +4 + 2(x-1)$

At F, $x = 1 \therefore F_F = +4$ kN

At E, $x = 2 \therefore F_E = +6$ kN

For section ED, $F_x = +4 + 2(x-1) + 6$

At E, $x = 2 \therefore F_E = +1$ kN. (Sudden change in S.F. due to point load)

At D, $x = 3 \therefore F_D = +14$ kN

For section DC, $F_x = +14$ kN (Constant)

At D, $x = 3 \therefore F_D = +14$ kN

At C, $x = 6 \therefore F_C = +14 \text{ kN}$

For section CA, $F_x = +14 + 2 = 16 \text{ kN}$ (Constant)

At C, $x = 6 \therefore F_C = +16 \text{ kN}$ (Sudden change in S.F. due to point load)

At A, $x = 8 \therefore F_A = +16 \text{ kN}$

The S.F.D. is shown in Fig. 3.9 (b).

B.M.D.: For section BF, $M_x = 4.x$

At B, $x = 0 \therefore M_B = 0$.

At F, $x = 1 \therefore M_F = -4 \text{ kN-m}$

For section FE, $M_x = -4.x - \frac{2}{2}(x-1)^2$ (Parabolic)

At F, $x = 1 \therefore M_F = -4 \text{ kN-m}$

At E, $x = 2 \therefore M_E = -9 \text{ kN-m}$

For section ED, $M_x = -4.x - \frac{2}{2}(x-1)^2 - 6(x-2)$ (Parabolic)

At E, $x = 2 \therefore M_E = -9 \text{ kN-m}$

At D, $x = 3 \therefore M_D = -22 \text{ kN-m}$

For section DC, This section is out of the udl. Hence the udl is to be converted into equivalent point load. The total udl (2×2) is assumed to be applied at C.G. of the udl i.e. 2 m away from the free end as shown in Fig. 3.9 (d). To draw BMD for section DA Fig. 3.9 (d) is to be considered.

$M_x = -4.x - 4.(x-2) - 6(x-2)$ (Linear)

At D, $x = 3 \therefore M_D = -22 \text{ kN-m}$

At C, $x = 6 \therefore M_C = -64 \text{ kN-m}$

For section CA

$M_x = -4.x - 4.(x-2) - 6.(x-2) - 2.(x-6)$ (Parabolic)

At C, $x = 6 \therefore M_C = -64 \text{ kN-m}$

At A, $x = 8 \therefore M_A = -96 \text{ kN-m}$

The B.M.D. is shown in Fig. 3.8 (c).

3.9 Shear Force And Bending Moment Diagrams For A Simply Supported Beam

3.9.1 A Simply Supported Beam With A Point Load At Mid-Point

Fig. 3.10 (a) shows a beam AB of length L simply supported at the ends A and B and carrying a point load W at its middle point C. Since the beam is loaded symmetrically the reactions at the support will be equal to $\frac{W}{2}$.

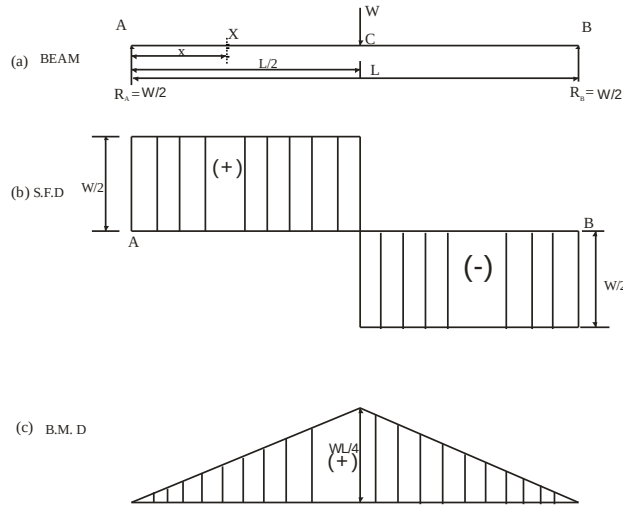


Fig. 3.10

$$\text{Hence } R_A = R_B = \frac{W}{2}.$$

S.F.D.: For the section AC let us take a section X at a distance x from the end A between A and C. Here we have considered the left portion of the section. The shear force at X will be equal to the resultant force acting on the left portion of the section. But the resultant force on the left portion is $\frac{W}{2}$ acting upwards. According to the sign convention, the resultant force on the left portion acting upwards is considered positive. Hence shear force at X is positive and its magnitude is $\frac{W}{2}$.

$$\therefore F_x = + \frac{W}{2} \text{ (Constant)}$$

For the section CB let us take a section X at a distance x from the end A. The resultant

force on the left portion will be

$$\left(\frac{W}{2} - W\right) = -\frac{W}{2}$$

This force will also remain constant between C and B. Hence shear force between C and B is equal to $-\frac{W}{2}$.

At the section C the shear force changes from $+\frac{W}{2}$ to $-\frac{W}{2}$.

The shear force diagram is shown in Fig. 3.10 (b).

B.M. D:

Section AC, $M_x = R_A \cdot x$

$$\text{or } M_x = +\frac{W}{2} \cdot x \text{ (Linear)} \quad \dots(i)$$

B.M. will be positive as for the left portion of the section, the moment of all forces at X is clockwise. Moreover, the bending of beam takes place in such a manner that concavity is at the top of the beam.

$$\text{At A, } x = 0 \quad \text{Hence } M_A = \frac{W}{2} \times 0 = 0$$

$$\text{At C, } x = \frac{L}{2} \quad \text{hence } M_C = \frac{W}{2} \times \frac{L}{2} = \frac{W \times L}{4}$$

$$\text{Section CB, } M_x = R_A \cdot x - W \times \left(x - \frac{L}{2}\right)$$

$$= \frac{W}{2} \cdot x - Wx + W \times \frac{L}{2} = \frac{WL}{2} - \frac{Wx}{2} \text{ (Linear)}$$

$$\text{At C, } x = \frac{L}{2} \quad \therefore M_C = \frac{WL}{2} - \frac{W}{2} \times \frac{L}{2} = \frac{W \times L}{4}$$

$$\text{At B, } x = L \quad \therefore M_B = \frac{WL}{2} - \frac{W}{2} \times L = 0$$

The B.M. diagram can be completed as shown in Fig. 3.10 (c).

It is to be worth noted the bending moment is maximum at the middle point C, where the shear force changes its sign.

3.9.2 A Simply Supported Beam With An Eccentric Point Load

Let us first calculate the reactions, by taking moments about A or about B.

Taking moments of the forces on the beam about A, we get

$$R_B \times L = W \times a$$

$$\therefore R_B = \frac{W.a}{L}$$

and

$$R_A = W - R_B = W - \frac{W.a}{L}$$

$$= W \left(1 - \frac{a}{L} \right) = W \left(\frac{L-a}{L} \right) = \frac{W \times b}{L} \quad (\because L = a + b)$$

S.F.D.: For the section AC let us consider a section X at distance x from the end A. The shear force F_x at the section is given by,

$$F_x = +R_A = + \frac{W.b}{L} \quad (\text{Constant})$$

The shear force will be positive as the resultant force on the left portion of the section is acting upwards.

For the section CB let us consider a section X at a distance x from the end A. The resultant force on the left portion will be $R_A - W$.

or

$$\frac{W.a}{L} - W = W \left(\frac{b-L}{L} \right) = -W \left(\frac{L-b}{L} \right) = - \frac{W.a}{L} \quad (\text{Constant})$$

At the section C, the shear force changes from $\frac{W.b}{L}$ to $-\frac{W.a}{L}$. The shear force diagram is shown in Fig 3.11 (b).

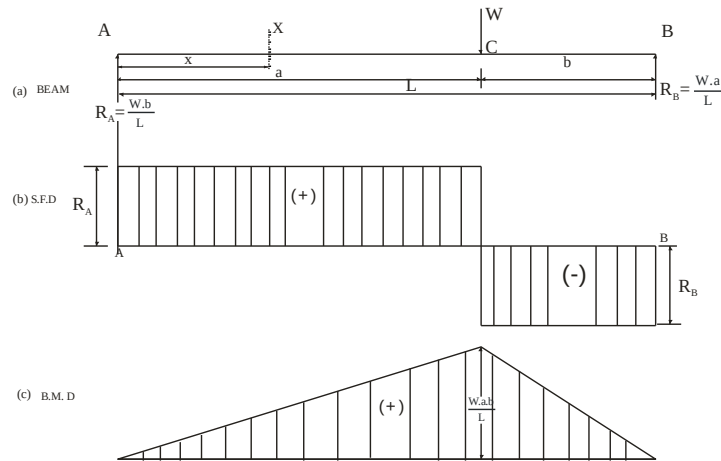


Fig. 3.11

B.M.D.

Section AC, $M_x = R_A \cdot x$

$$= + \frac{W.b}{L} \cdot x \text{ (Linear) (Plus sign due to sagging)}$$

$$\text{At A, } x = 0 \quad \text{Hence } M_A = \frac{W.b}{L} \times 0 = 0$$

$$\text{At C, } x = a \quad \text{Hence } M_C = \frac{W.b}{L} \cdot a = \frac{W \cdot a \cdot b}{L}$$

The B.M. is zero at B. Hence B.M. will decrease from $\frac{W \cdot a \cdot b}{L}$ at C to zero at B following a straight line law. The B.M. diagram is drawn in Fig. 3.11 (c).

It is to be worth noted from the shear force and bending moment diagrams, the B.M. is maximum at C where the S.F. changes its sign.

3.9.3 A Simply Supported Beam Carrying A Uniformly Distributed Load

Fig. 3.12 shows a beam AB of length L simply supported at the ends A and B and carrying a uniformly distributed load of w per unit length over the entire length. The reactions at the supports will be equal and their magnitude will be half the total load on the entire length.

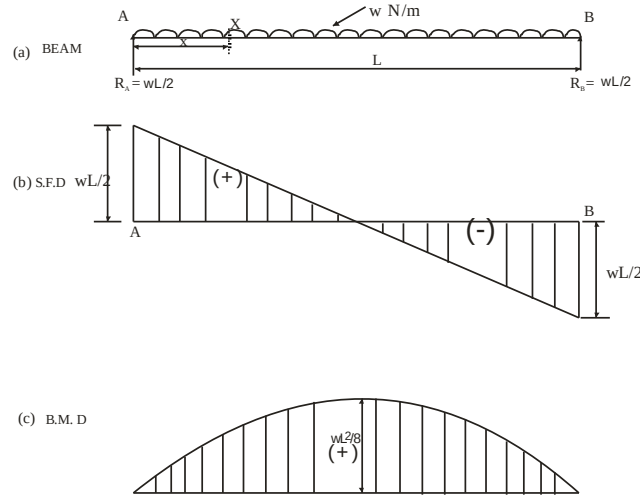


Fig. 3.12

$$\therefore R_A = R_B = \frac{wL}{2}$$

S.F.D.: For the section AC let us consider a section X at distance x from the end A. The shear force F_x at the section is given by,

$$F_x = + R_A - w \cdot x = + \frac{wL}{2} - w \cdot x \text{ (Linear)} \quad \dots(i)$$

The values of shear force at different points are :

$$\text{At A, } x = 0 \quad \therefore F_A = + \frac{wL}{2} - \frac{w \cdot 0}{2} = + \frac{wL}{2}$$

$$\text{At B, } x = L \quad \therefore F_B = + \frac{wL}{2} - w \cdot L = - \frac{wL}{2}$$

$$\text{At C, } x = \frac{L}{2} \quad \therefore F_C = + \frac{wL}{2} - w \cdot \frac{L}{2} = 0$$

The shear force diagram is drawn as shown in Fig. 3.12 (b).

B.M.D.

Section AC, $M_x = + R_A \cdot x - w \cdot x \cdot \frac{x}{2}$

$$= \frac{w.L}{2} \cdot x - \frac{w.x^2}{2} \quad \left(\because R_A = \frac{w.L}{2} \right) \quad \text{.....(ii)}$$

(Parabolic)

The values of B.M. at different points are :

$$\text{At A, } x = 0 \quad \therefore M_A = \frac{w.L}{2} \cdot 0 - \frac{w.0}{2} = 0$$

$$\text{At B, } x = L \quad \therefore M_B = \frac{w.L}{2} \cdot L - \frac{w}{2} \cdot L^2 = 0$$

$$\begin{aligned} \text{At C, } x = \frac{L}{2} \quad \therefore M_C &= \frac{w.L}{2} \cdot \frac{L}{2} - \frac{w}{2} \cdot \left(\frac{L}{2} \right)^2 \\ &= \frac{w.L^2}{4} - \frac{w.L^2}{8} = + \frac{w.L^2}{8} \end{aligned}$$

Thus the B.M. increases according to parabolic law from zero at A to $+\frac{w.L^2}{8}$ at the middle point of the beam and from this value the B.M. decreases to zero at B according to the parabolic law.

The B.M. diagram is drawn as shown in Fig. 3.12(c).

3.9.4 A Simply Supported Beam With Combination of Loads (U.D.L. And Point loads)

Let us first calculate the reactions, by taking moments about A or about B.

Let us take moments of the forces on the beam about A.

$$R_B \times 8 = 6 \times 6 + 2 \times 2 \times 2 \times 6$$

($2 \times 2 \times 6$ is the moment of udl about support A. The udl is converted into point load (Fig. 3.12 (d)) and applied at C.G. of the udl.)

$$\therefore R_B = 8 \text{ kN}$$

Similarly taking moments of the forces on the beam about B.

$$R_A \times 8 = 2 \times 6 + 6 \times 2 - 2 \times 2 \times 2$$

($2 \times 2 \times 2$ is the moment of udl about support B. The udl is converted into point load (Fig. 3.13 (d)) and applied at C.G. of the udl.)

$$\therefore R_A = 4 \text{ kN}$$

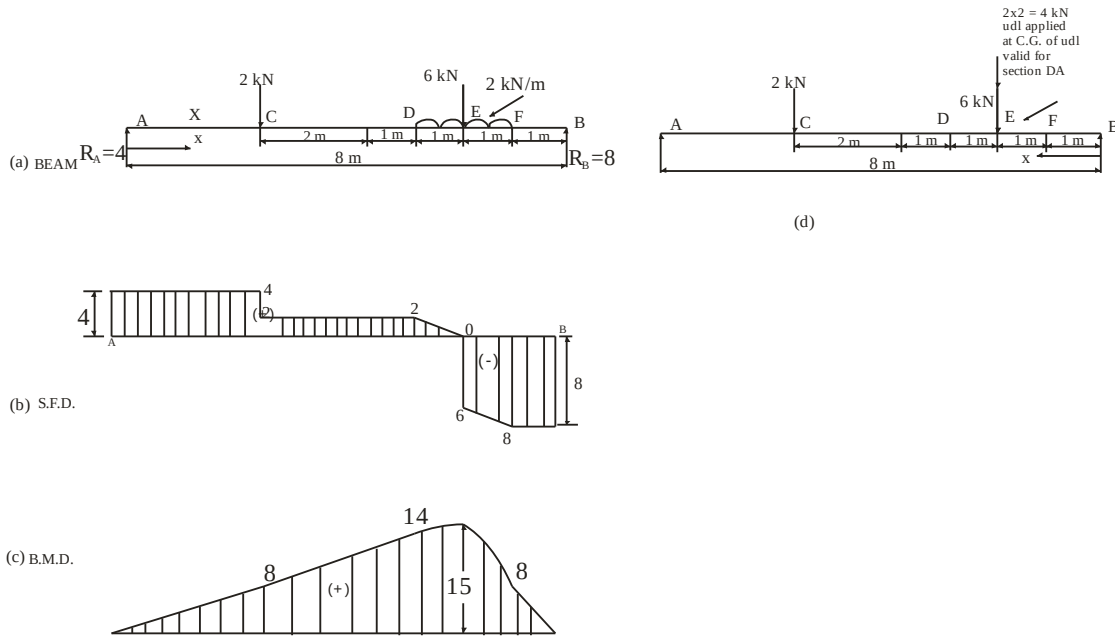


Fig. 3.13

S.F.D.: For the section AC let us consider a section X at distance x from the end A. The shear force F_x at the section is given by,

$$F_x = +R_A = +4 \text{ kN (Constant)}$$

The shear force will be positive as the resultant force on the left portion of the section is acting upwards.

Section CD, $F_x = +4 - 2 = 2 \text{ kN (Constant)}$

Section DE, $F_x = +4 - 2 - 2(x-5) \text{ (Linear)}$

At D, $x = 5 \therefore F_D = 2 \text{ kN}$

At E, $x = 6 \therefore F_E = 0 \text{ kN}$

Section EF, $F_x = +4 - 2 - 2(x-5) - 6 \text{ (Linear)}$

At E, $x = 6 \therefore F_E = -6 \text{ kN}$ (Sudden change due to point load)

At F, $x = 7 \therefore F_F = 8 \text{ kN}$ (constant)

Section FB, $F_x = +4 - 2 - 2x - 6$
 $= 8 \text{ kN (Constant)}$

At F, $x = 7 \therefore F_F = -8 \text{ kN}$

At B, $x = 8 \therefore F_B = -8 \text{ kN}$

The shear force diagram is shown in Fig 3.13 (b).

B.M.D.

Section AC, $M_x = R_A \cdot x$ (Linear)

$= +4 \cdot x$ (Linear) (Plus sign due to sagging)

At A, $x = 0 \therefore M_A = 4 \times 0 = 0$

At C, $x = 2 \therefore M_C = 4 \times 2 = 8 \text{ kN-m}$

Section CD, $M_x = +4 \cdot x - 2 \cdot (x-2)$ (Linear)

At C, $x = 2 \therefore M_C = 8 \text{ kN-m}$

At D, $x = 5 \therefore M_D = 14 \text{ kN-m}$

Section DE, $M_x = +4 \cdot x - 2 \cdot (x-2) - \frac{2}{2}(x-5)^2$ (Parabolic)

At D, $x = 5 \therefore M_D = 14 \text{ kN-m}$

At E, $x = 6 \therefore M_E = 15 \text{ kN-m}$

Section EF, $M_x = +4 \cdot x - 2 \cdot (x-2) - \frac{2}{2}(x-5)^2 - 6(x-6)$ (Parabolic)

At E, $x = 6 \therefore M_E = 15 \text{ kN-m}$

At F, $x = 7 \therefore M_F = 8 \text{ kN-m}$

For section FB, This section is out of the udl. Hence the udl is to be converted into equivalent point load. The total udl (2×2) is assumed to be applied at C.G. of the udl i.e. 2 m away from the

support B as shown in Fig. 3.13 (d). To draw BMD for section FB Fig. 3.13 (d) is to be considered.

$$\text{Section FB, } M_x = +4 \cdot x - 2 \cdot (x-2) - 4 \cdot (x-6) - 6 \cdot (x-6) \text{ (Linear)}$$

$$\text{At F, } x = 7 \quad \therefore M_F = 8 \text{ kN-m}$$

$$\text{At B, } x = 8 \quad \therefore M_B = 0$$

The B.M. is zero at B. The B.M. diagram is drawn in Fig. 3.13 (c).

It is to be worth noted from the shear force and bending moment diagrams, the B.M. is maximum at E where the S.F. changes its sign.

3.10 Shear Force And Bending Moment Diagrams For Over-Hanging Beams

If the end portion of a beam is extended beyond the support such beam is known as overhanging beam. In case of overhanging beams, the B.M. is positive between the two supports, whereas the B.M. is negative for the over-hanging portion. Hence at some point, the B.M. is zero after changing its sign from positive to negative or vice-versa. That point is known as the point of contraflexure or point of inflexion.

3.10.1 Beam With Both Sides Overhang: Point Loads

Let us first calculate the reactions, by taking moments about A or about B.

Let us take moments of the forces on the beam about B.

$$8 \times 5 = R_A \times 4 + 10 \times 2$$

$$\therefore R_A = 5 \text{ kN}$$

$$\therefore R_B = 8 + 10 - 5 = 13 \text{ kN}$$

S.F.D.: For the section CA,

$$F_x = - 8 \text{ kN (Constant)}$$

The shear force will be negative as the resultant force on the left portion of the section is acting downwards.

$$\text{Section AB,} \quad F_x = - 8 + 5 = - 3 \text{ kN (Constant)}$$

$$\text{Section BD,} \quad F_x = - 8 + 5 + 13 = +10 \text{ kN}$$

Thus S.F. changes sign at B.

The shear force diagram is shown in Fig 3.14 (b).

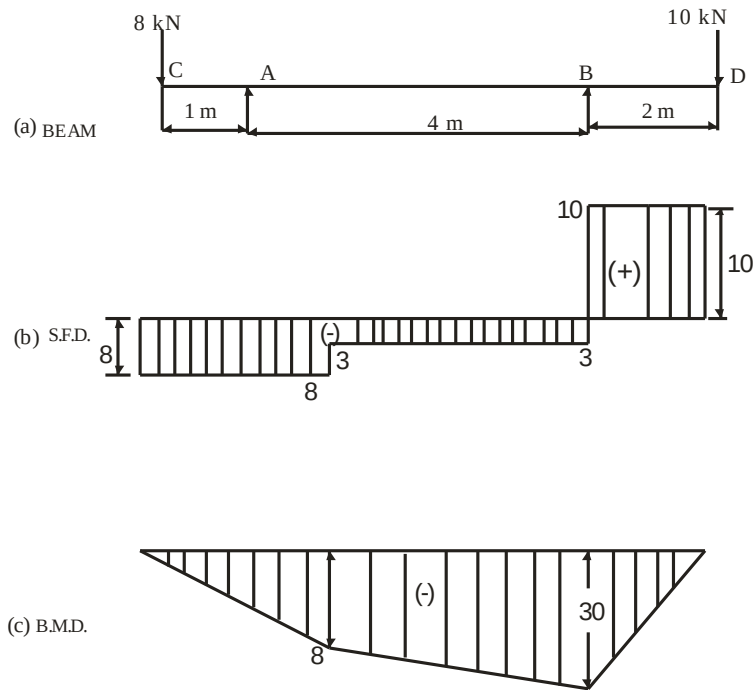


Fig. 3.14

B.M.D.

Section CA, $M_x = -8 \cdot x$ (Linear)

$$\text{At C, } x = 0 \quad \therefore M_C = -8 \times 0 = 0$$

$$\text{At A, } x = 1 \quad \therefore M_C = -8 \times 1 = -8 \text{ kN-m}$$

Section AB, $M_x = -8 \cdot x + 5 \cdot (x-1)$ (Linear)

$$\text{At A, } x = 1 \quad \therefore M_A = -8 \text{ kN-m}$$

$$\text{At B, } x = 5 \quad \therefore M_B = -20 \text{ kN-m}$$

Section BD, $M_x = -8 \cdot x + 5 \cdot (x-1) + 13 \cdot (x-5)$ (Linear)

$$\text{At B, } x = 5 \quad \therefore M_B = -20 \text{ kN-m}$$

$$\text{At D, } x = 7 \quad \therefore M_D = 0$$

The B.M. diagram is drawn in Fig. 3.14 (c). Maximum bending moment is where S.F. changes sign.

3.10.2 Beam With Both Sides Overhang: Uniformly Distributed Load

Fig. 3.15 (a) shows a beam EABD of length $(l+2a)$ with supports at A and B. The beam carries a udl w N/m over entire length.

Since the beam is symmetrically loaded the reactions R_A and R_B are

$$R_A = R_B = \frac{w}{2}(l + 2a)$$

S.F.D.: For the section EA,

$$F_x = -wx \text{ (Linear)}$$

The shear force will be negative as the resultant force on the left portion of the section is acting downwards.

$$\text{S.F. at E, } x = 0 \therefore F_E = 0$$

$$\text{S.F. at A, } x = a \therefore F_A = -w.a$$

For the section AB,

$$F_x = R_A - wx = \frac{w}{2}(l + 2a) - wx \text{ (Linear)}$$

$$\text{S.F. at A, } x = a \therefore F_A = + \frac{wl}{2} \text{ (Sudden change in S.F. due to point load } R_A)$$

$$F_x = R_A - wx = \frac{w}{2}(l + 2a) - wx \text{ (Linear)}$$

$$\text{S.F. at B, } x = (a+l) \therefore F_B = - \frac{wl}{2}$$

Hence from A to B the S.F. changes uniformly from $+\frac{wl}{2}$ to $-\frac{wl}{2}$. The S.F. at C the middle point of the beam is zero.

For the section BD, the S.F. will change uniformly from $+wa$ at B to zero at D.

The shear force diagram is shown in Fig 3.15 (b).

B.M.D.

Section EA, $M_x = -\frac{wx^2}{2}$ (Hogging and parabolic)

$$\text{At E, } x = 0 \quad \therefore M_E = 0$$

$$\text{At A, } x = a \quad \therefore M_A = -\frac{wa^2}{2}$$

$$\text{Section AB, } M_x = R_a(x-a) - \frac{wx^2}{2} = \frac{w}{2}(l+2a)(x-a) - \frac{wx^2}{2}$$

$$\text{At A, } x = a \quad \therefore M_A = -\frac{wa^2}{2}$$

$$\text{At C, } x = a + \frac{l}{2} \quad \therefore M_C = \frac{w(l^2 - 4a^2)}{8}$$

Case (A): when $l > 2a$

For this case M_c is positive.

For this case there will be two points of contraflexure C_1 and C_2 between A and B. The positions of these points can be determined by equating the general expression for B.M. for any section in AB to zero.

$$M_x = \frac{w}{2}(l+2a)(x-a) - \frac{wx^2}{2}$$

For the points of contraflexure

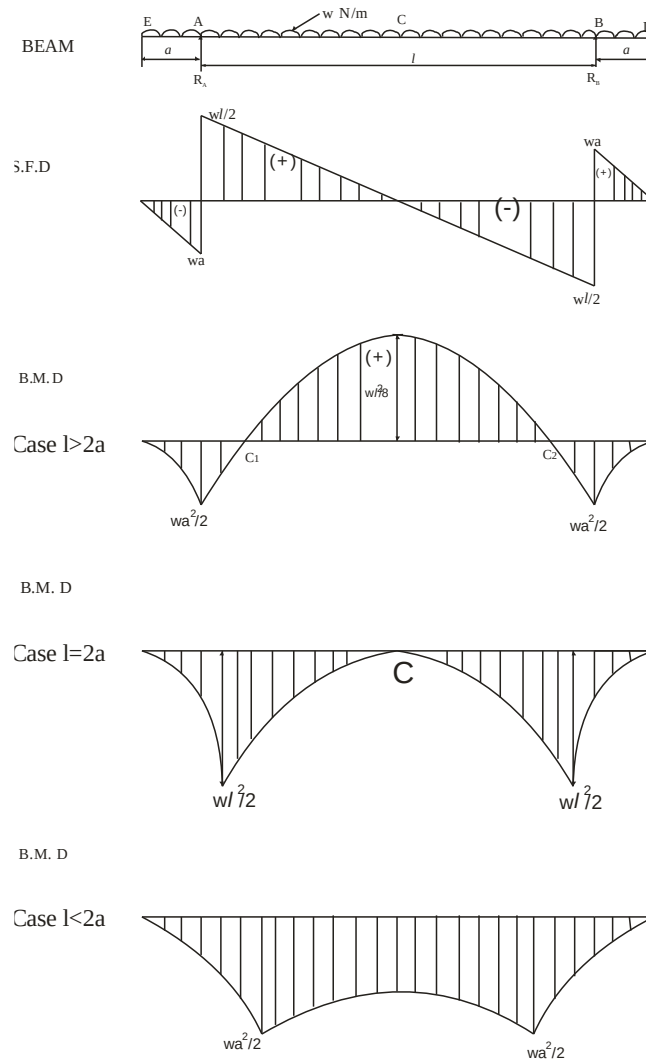


Fig. 3.15

$$M_x = \frac{w}{2}(l+2a)(x-a) - \frac{wx^2}{2} = 0$$

$$\text{Or } \frac{w}{2}(l+2a)(x-a) - \frac{wx^2}{2} = 0$$

$$\text{Or } x^2 - (2a+l)x + a(2a+l) = 0$$

$$x = \frac{(2a + l) \pm \sqrt{(2a + l)^2 - 4a(2a + l)}}{2}$$

$$\text{or } x = a + \frac{l}{2} \pm \frac{\sqrt{(l)^2 - 4a^2}}{2}$$

$$\text{Here EC} = a + \frac{l}{2}$$

Hence the points of contraflexure C_1 and C_2 are at a distance $\frac{\sqrt{(l)^2 - 4a^2}}{2}$ from the middle point C of the beam. The BMD is as shown in Fig. 3.15 (c).

Case (B): when $l=2a$ B.M. at C = 0

The points of contraflexure will coincide with C and the BMD between A and B just touches the span. The BMD is as shown in Fig. 3.15 (d).

Case (C): when $l < 2a$

$$M_C = \frac{w(l^2 - 4a^2)}{8} \quad \text{since } 4a^2 > l^2 \therefore M_C \text{ is negative.}$$

The BMD is as shown in Fig. 3.15 (e).

Chapter 3

Problem 3.1 A cantilever AB of length l is subjected to a downward load P at its free end and an upward load P at a distance a from the free end. Draw S.F. and B.M. diagrams for the cantilever.

Solution:

S.F.D.

Section in BC = $+W$

Section in CA = 0

B.M. at any section in BC = $-W.x$ (Linear);

At B $x = 0 \therefore M_B = 0$

At C $x = a \therefore M_C = -Wa$

B.M. at any section in CA = $-W.x + W.(x-a)$

a)

$= -Wa$ (Constant)

At C $x = a \therefore M_C = -Wa$

At A $x = l \therefore M_A = -Wa$

From C to D the

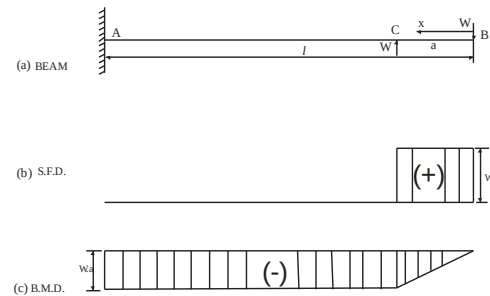


Fig. 3.16

Problem 3.2 A cantilever beam 5 m long carries concentrated loads of 30, 30 and 30 kN at distance of 1.5, 3 and 4.5 meters from the fixed end. In addition to this the beam carries a uniformly distributed load of 10 kN/m over the entire length of the beam. Draw the S.F. and B.M. diagrams.

Solution :

The loaded beam is shown in Fig. 3.17

S.F. at B = 0

S.F. at C = $(10 \times 0.5 + 30) = 35$ kN

S.F. at D = $(10 \times 2 + 30 + 30) = 80$ kN

S.F. at E = $(10 \times 3.5 + 30 + 30 + 30) = 125$ kN

S.F. at A = $(10 \times 5 + 30 + 30 + 30) = 140$ kN

The S.F. diagram is shown in Fig. 3.17(b)

B.M. at B = 0

B.M. at C = $-10 \times 0.5 \times \frac{0.5}{2} = -1.25$ kN.m

B.M. at D = $-10 \times 2 \times \frac{2}{2} - 30 \times 1.5 = -65$

kN.m

B.M. at E = $-10 \times 3.5 \times \frac{3.5}{2} - 30 \times 3 - 30 \times 1.5 = -196.25$ kN.m

B.M. at A = $-10 \times 5 \times \frac{5}{2} - 30 \times 4.5 - 30 \times 3 - 30 \times 1.5$

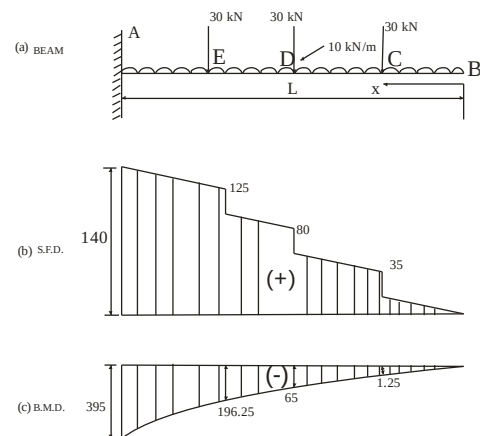


Fig. 3.17

$$= -395 \text{ kN.m}$$

The B.M. diagram is shown in Fig. 3.17(c)

Problem 3.3 Draw the shear force and bending moment diagrams for the beam shown in Fig. 3.18. Also find the position and magnitude of the maximum bending moment.

Solution: Let the salient point of the beam be named as shown in the Fig. 3.18. Let R_a and R_b be the vertical reactions at the left and right supports respectively.

Taking moments about the left support, we have

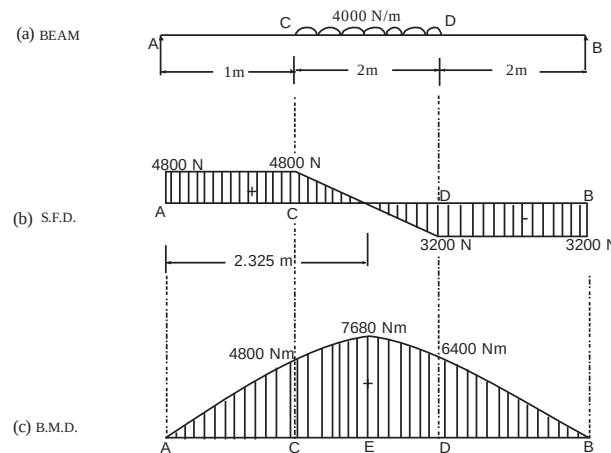


Fig. 3.18

$$\begin{aligned} R_b \times 5 &= 4000 \times 2(1+1) \\ \therefore R_b &= 3200 \text{ N} \\ \therefore R_a &= (4000 \times 2) - 3200 \\ &= 4800 \text{ N} \end{aligned}$$

S.F. between A and C = +4800 N

S.F. between D and B = - 3200 N

Let the S.F. be zero at a section E, x meters from A

Equating the S.F. at E to zero, we have

$$\begin{aligned} 4800 - 4000(x - 1) &= 0 \\ \therefore x &= 2.2 \text{ m} \end{aligned}$$

B.M. calculations

B.M. at A = $M_a = 0$

B.M. at C = $M_c = 4800 \times 1 = 4800 \text{ Nm}$

B.M. at D = $M_d = 3200 \times 2 = 6400 \text{ Nm}$

B.M. at 2.2 m from A

$$\begin{aligned} &= M_E = 4800 \times 2.2 - 4000 \frac{(2.2 - 1)^2}{2} \text{ Nm} \\ &= 7680 \text{ Nm} \end{aligned}$$

Problem 3.4 Draw S.F. and B.M. diagrams for the simply supported beam shown in Fig. 3.19.

Solution: Let R_a and R_b be the vertical reactions at the supports A and B respectively. For the equilibrium of the beam taking moments of the forces on the beam about the left support, we have,

$$\begin{aligned} R_b \times 8 &= 4 \times 1.5 + 10 \times 4 + 3 \times 6 = 64 \text{ kN/m} \\ \therefore R_b &= \frac{64}{8} = 8 \text{ kN} \end{aligned}$$

$\therefore R_a = \text{total load on the beam} - R_b = 17 - 8 = 9 \text{ kN}$
 S.F. between A and C = 9 kN
 S.F. between C and D = $+9 - 4 = +5 \text{ kN}$
 S.F. between D and E = $+9 - 4 - 10 = -5 \text{ kN}$
 or alternatively = $-8 + 3 = -5 \text{ kN}$
 S.F. between E and b = $+9 - 4 - 10 - 3 = -8 \text{ kN}$
 or alternatively = $-R_b = -8 \text{ kN}$

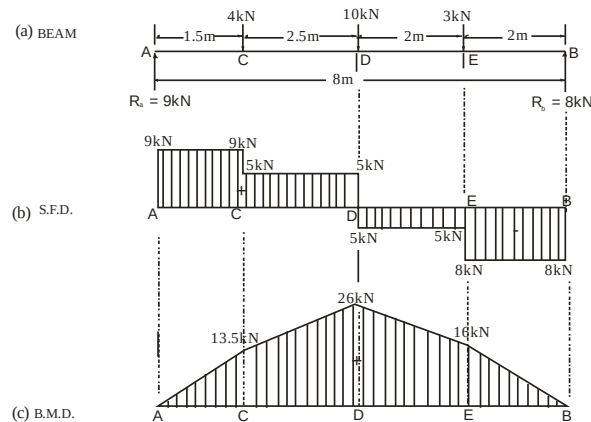


Fig. 3.19

B.M. at A = 0
 B.M. at C = $+9 \times 1.5 = +13.5 \text{ kNm}$ (sagging)
 B.M. at D = $+9 \times 4 - 4 \times 2.5 = +26 \text{ kNm}$ (Sagging)
 B.M. at E = $+8 \times 2 = +16 \text{ kNm}$ (Sagging)

It may be observed from the S.F. and B.M. diagrams that the maximum B.M. occurs at D where the S.F. change its sign. SFD and BMD are shown in Fig. 3.19.

Problem 3.5 A beam AB 10 meters long has supports at its ends A and B. It carries a point load of 10 kN at 3 meters from A and a point load of 5 kN at 7 metres from A and a uniformly distributed load of 1 kN per meter between the point loads. Draw SF and BM diagrams for the beam.

Solution:

Reactions :

Since the loading is symmetrical reaction at each support equals half the total load

$$\therefore R_a = R_b = \frac{10 + 10 + 2 \times 4}{2} = 14 \text{ kN}$$

S.F. analysis

S.F. at any section in AD = + 14 kN
 S.F. just on RHS of D = $+14 - 10 = +4 \text{ kN}$
 S.F. just on LHS of E = $-14 + 10 = -4 \text{ kN}$
 S.F. at any section in EB = - 14 kN
 S.F. at centre C = 0

B.M. analysis

B.M. at A = BM at B = 0

$$\text{B.M. at D} = \text{BM at E} = + 14 \times \frac{2}{2} \times 3 = + 42 \text{ kNm}$$

$$\text{B.M. at C} = 14 \times 5 - 10 \times 2 - 2 \times \frac{2}{2} = 70 - 20 - 4 = + 46 \text{ kNm}$$

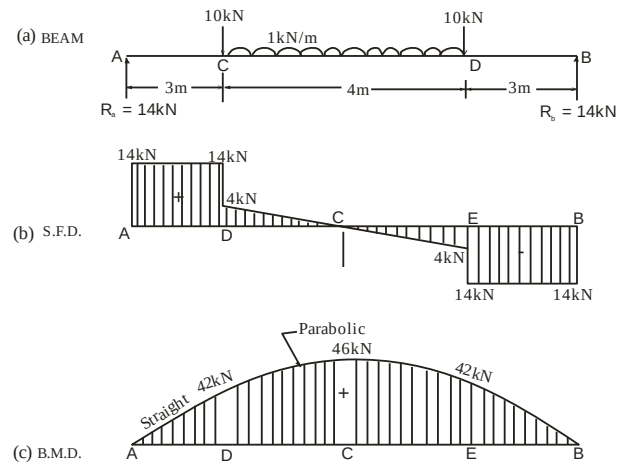


Fig. 3.20

(Maximum bending moment)

The B.M. diagram is linear for the parts AD and BE and is parabolic for the part DE. SFD and BMD are shown in Fig. 3.20.

Problem 3.6 A simply supported beam is loaded as shown in Fig. 3.21. Draw the S.F. and B.M. diagrams and find the value of maximum bending moment.

Solution: Let R_A and R_B be the reactions at the supports A and B respectively.

$$R_A + R_B = 30 \times 2 + 120 + 50 \times 4 = 380 \text{ kN} \quad \dots(i)$$

Taking moment about A

$$R_B \times 8 = 30 \times 2 \times \left(\frac{2}{2} + 6 \right) + 120 \times 6 + 50 \times 4 \times \frac{4}{2} = 1540$$

$$\therefore R_B = \frac{1540}{8} = 192.5 \text{ kN}$$

Putting the value of R_B in Equation (i) we get $R_A = 380 - 192.5 = 187.5 \text{ kN}$

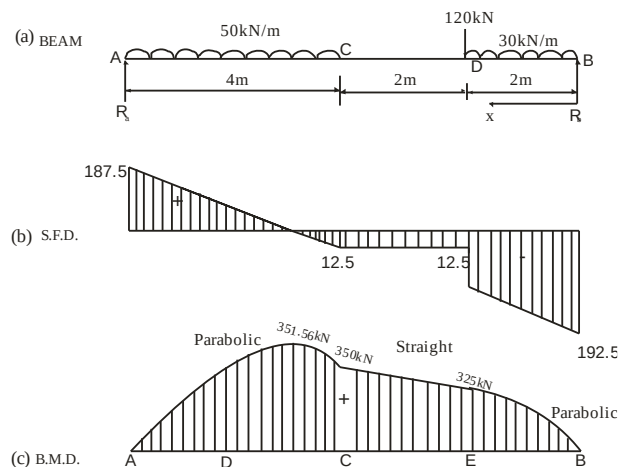


Fig. 3.21

$$\text{S.F. at B} = 192.5 \text{ kN (-) ve}$$

$$\text{S.F. at C} = 192.5 - 30 \times 2 - 120 = 12.5 \text{ kN (-) ve}$$

$$\text{S.F. at D} = 12.5 \text{ kN (-) ve}$$

S.F. at A = 187.5 kN (+) ve

The S.F. diagram is shown in Fig. 3.21 There is a point between A and D where S.F. is zero. Let the point be E and BE = x

Now S.F. at E = $-192.5 + 30 \times 2 + 120 + 50(x - 4)$

$$0 = -12.5 + 50x + 200$$

or $50x = 212.5$

$\therefore x = 4.25 \text{ m}$

B.M. at A = 0

B.M. at B = 0

$$\text{B.M. at C} = 192.5 \times 2 - 30 \times 2 \times \frac{2}{2} = 325 \text{ kN.m}$$

$$\text{B.M. at D} = 192.5 \times 4 - 30 \times 2 \times \left(\frac{2}{2} + 2\right) - 120 \times 2 = 350 \text{ kN.m}$$

$$\begin{aligned} \text{B.M. at E} &= 192.5 \times 4.25 - 30 \times 2 \times (4.25 - 1) - 120 \times 2.25 \\ &\quad - 50 \times 0.25 \times \frac{2}{2} = 351.56 \text{ kN.m} \end{aligned}$$

Therefore the maximum bending moment is at E and is equal to 351.56 kN.m.

The B.M. diagram is shown in Fig. 3.21.

Problem 3.7 A simply supported beam ABC with supports at A and B, 6 meters apart and with an overhang BC 2 meters long carries a uniformly distributed load of 30 kN per meter over the whole length as shown in Fig. 3.22 Draw S.F. and B.M. diagrams.

Solution:

Reactions. Taking moments about the end A, $\frac{8}{2}$

$$R_b \times 6 = 30 \times 8 \times \frac{2}{2}$$

$\therefore R_b = 160 \text{ kN}$

$\therefore R_a = \text{Total load} - R_b = (30 \times 8) - 160 = 80 \text{ kN}$

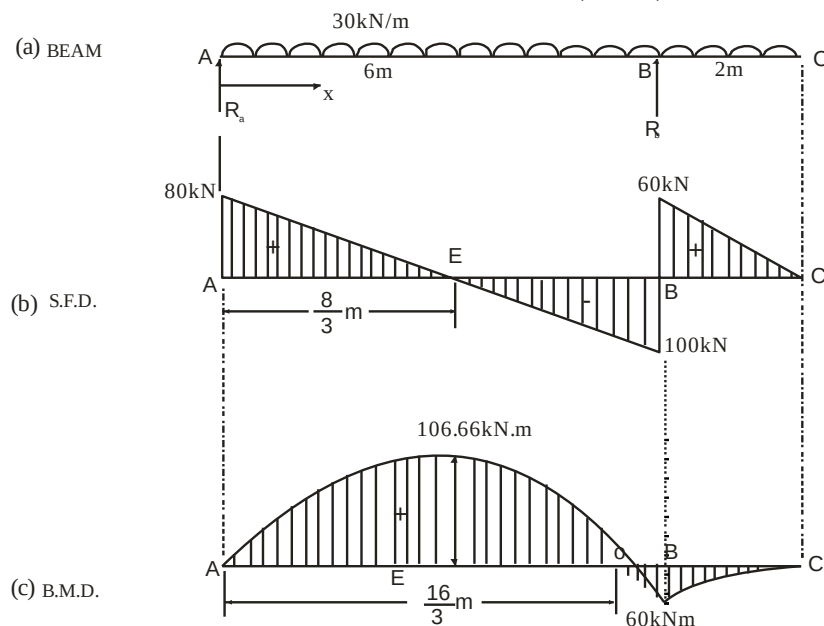


Fig. 3.22

S.F.D.: At any section AB distant x from A,

Shear force $S = 80 - 30x$

At $x = 0$, $S = +80$ kN

At $x = 6$ m, $S = 80 - 30 \times 6 = -100$ kN

Section of zero shear: Equating the general expression for shear force to zero.

$$80 - 30x = 0$$

$$\therefore x = \frac{80}{30} \text{ m} = 2.67 \text{ m}$$

At any section in CB, distant x from C,

Shear force $S = +30x$

At $x = 0$, $S = 0$

At $x = 2$ m, $S = 30 \times 2 = +60$ kN

B.M.D.: At any section in AB, distant x from A, $\frac{x^2}{2}$

Bending moment $= M = 80x - 30 \times \frac{x^2}{2}$

$$\therefore M = 80x - 15x^2$$

This is a parabolic law.

At $x = 0$, $M = 0$

At $x = 6$ m, $M = 80 \times 6 - 15 \times 6^2 = -60$ kNm

Maximum bending moment which occurs at the section of zero shear

$$\begin{aligned} &= M_e = 80 \times \frac{8}{3} - 15 \left(\frac{8}{3} \right)^2 \\ &= + \frac{320}{3} \text{ kNm} = +106.66 \text{ kNm} \end{aligned}$$

At any section in CB, distant x and C, $\frac{x^2}{2}$

Bending moment $= M = -30 \frac{x^2}{2}$

$$\therefore M = -15x^2 \text{ . this is a parabolic law.}$$

At $x = 0$, $M = 0$

At $x = 2$ m, $M = -15 \times 2^2 = -60$ kNm

Problem 3.8 Draw shear force and bending Moment diagrams for the beam shown in

Fig. 3.23.

Solution : Taking moments about A, $R_b \times 10 = 20 \times 14(7-3)$

$$\therefore R_b = 112 \text{ KN} \uparrow \quad \therefore R_a = (20 \times 14) - 112 = 168 \text{ KN} \uparrow$$

S.F.D.: S.F. at C = 0,

S.F. on LHS of A = $-20 \times 3 = -60$ kN

S.F. on RHS of A = $-60 + 168 = +108$ kN;

S.F. on LHS of B = $168 - 20 \times 13 = -82$ kN

S.F. on RHS of B = $-82 + 112 = +20$ kN ;

S.F. at D = 0

Section of zero shear. Let the shear force be zero at a section distant x from A.

Equating the shear force to zero,

$$168 - 20(3+x) = 0 \quad \therefore x+3 = 8.4 \quad \therefore x = 5.4 \text{ m}$$

B.M.D. BM at C = 0

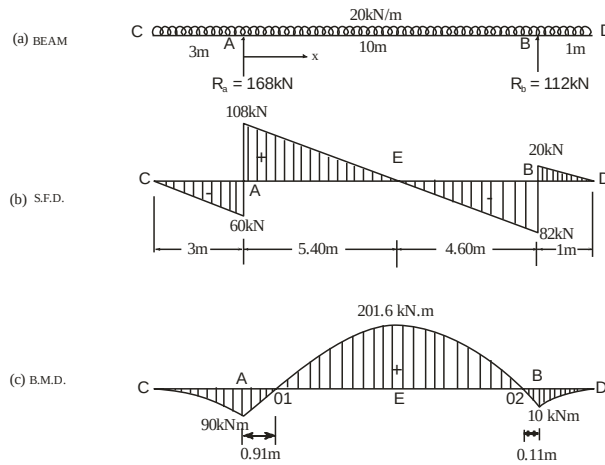


Fig. 3.23

$$\text{B.M. at A} = -\frac{20 \times 3^2}{2} = -90 \text{ kNm};$$

$$\text{B.M. at B} = -\frac{20 \times 1^2}{2} = -10 \text{ kNm}$$

B.M. at the section of zero shear, i.e. at a distance of 5.40 m from A,

$$M_{\max} = 168 \times 5.4 - \frac{20 \times 5.4^2}{2} = 201.6 \text{ kNm}$$

Points of contra flexure between A and B. Let the B.M. be zero at a distance x from A. Equating the general expression for B.M. to zero, $168x - \frac{20(x^2)}{2} = 0$

$$\therefore x^2 - 10x + 9 = 0 \text{ solving we get,}$$

$$x = 0.91 \text{ m and } x = 9.89 \text{ m}$$

Distance of point of contra flexure 0_1 from A = 0.91 m

Distance of point of contra flexure 0_2 from B = $10 - 9.89 = 0.11 \text{ m}$

Problem 3.9 Calculate the reactions at the supports A and B of the beam shown in Fig. 3.24. Draw bending moment and shearing force diagrams.

Solution: Let reactions at A and B be R_a and R_b respectively.

Taking moments about A, We have, $R_b \times 7 + 1000 \times 2$

$$= 2500 + 1500 \times 10$$

$$R_b = \frac{23000}{7} = 3285.71$$

$$\therefore R_a = \text{Total load} - R_b = 5000 - 3285.71 = +1714.29$$

S.F. Calculations

S.F. between D and A = - 1000 N

S.F. between A and C = - 1000 + 1714.29 = + 714.29

S.F. between C and B = + 1500 - 3285.71 = - 1785.71

S.F. between B and E = + 1500 N

B.M. Calculations

B.M. at D = $M_d = 0$

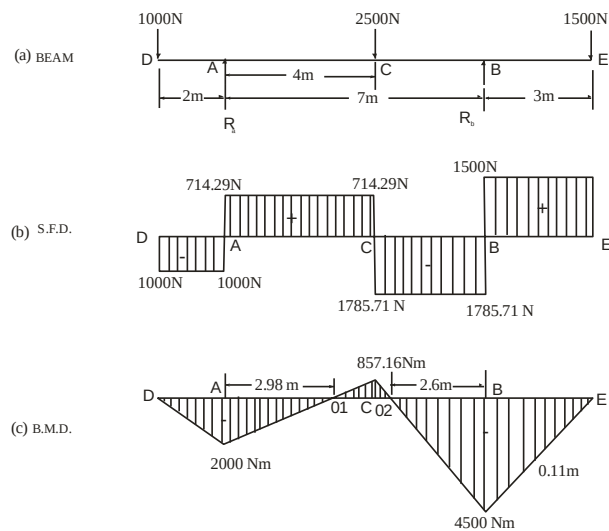


Fig. 3.24

$$\text{B.M. at A} = M_a = -1000 \times 2 = -2000 \text{ Nm}$$

$$\begin{aligned} \text{B.M. at C} = M_c &= +1714.29 \times 4 - 1000 \times 6 \text{ Nm} \\ &= -857.16 \end{aligned}$$

$$\text{B.M. at B} = M_b = -1500 \times 3 = -4500 \text{ Nm}$$

$$\text{B.M. at E} = M_e = 0$$

There will be two points of contra flexure O_1 and O_2 . One of them lies between A and C and the other lies between C and B.

Points of contra flexure between A and C: Let this point be x meters from A. Equating the general expression for B.M. to zero, $1714.29x - 1000(x+2) = 0$

Solving we get, $x = 2.8$ m from A.

Points of contra flexure between C and B: Let this point be x meters from B. Equating the general expression for B.M. to zero, $3285.71x - 1500(x+3) = 0$

Solving we get, $x = 2.52$ m from B.

Problem 3.10 A beam of uniform section 10 meters long carries a uniformly distributed load of 10 kN per meter over the whole length and a concentrated load of 10 kN at the right end. If the beam is freely supported at the left and, find the position of the second support so that maximum bending moment for the beam shall be as small as possible. Find also the maximum bending moment for this case.

Solution: Let the supports be at A and B so that the overhang BC = a meter

$\therefore$

$$AB = (10 - a) \text{ meter}$$

Taking moments about A, $R_b (10 - a) = \frac{10 \times 10 \times 10}{2} + 10 \times 10$

$$R_b = \left(\frac{600}{10 - a} \right) \text{ kN}$$

$\therefore$

$$R_a = (10 \times 10) + 10 - \frac{600}{10 - a} = \frac{500 - 110a}{10 - a} \text{ kN}$$

$$\text{B.M. at B} = -10 \times a - 10 \times \frac{a^2}{2} = -5a(a + 2)$$

Maximum positive B.M. will occur at a section in AB.

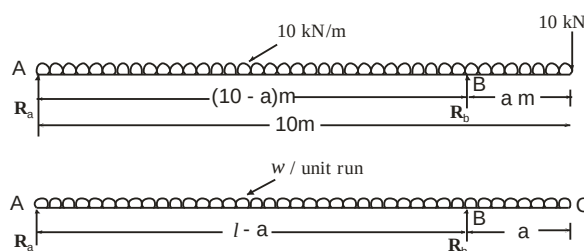


Fig. 3.24

Let this section be x meters from A.

$$\therefore \text{S.F. at this section} = 0$$

$$\therefore \frac{500 - 110a}{10 - a} - 10x = 0$$

$$\therefore x = \frac{500 - 110a}{(10 - a)100} = \frac{50 - 11a}{10 - a} m$$

$$\begin{aligned} \text{Maximum positive B.M.} &= R_a \cdot x - \frac{wx^2}{2} = \frac{500 - 110a}{10 - a} \cdot \frac{50 - 11a}{10 - a} - \frac{10}{2} \left[\frac{50 - 11a}{10 - a} \right]^2 \\ &= \frac{(50 - 11a)^2}{(10 - a)^2} \\ &= 5 \frac{(50 - 11a)^2}{(10 - a)^2} \end{aligned}$$

For the condition that the maximum bending moment shall be as possible, the maximum positive bending moment and the hogging moment over the support should be numerically equal.

$$\begin{aligned} \therefore \frac{(50 - 11a)^2}{5(10 - a)^2} &= 5a(a + 2) \\ \therefore \left[\frac{50 - 11a}{10 - a} \right]^2 &= a(a + 2) \end{aligned}$$

Solving by trial and error, we get $a = 2.33$ m.

Maximum B.M. with the above value of a

$$= 5a(a + 2) = 5 \times 2.33(2.33 + 2) = 50.44 \text{ kNm}$$

Problem 3.11 A beam AB, 20 meters long supported on two intermediate props 12 meters apart carries a uniformly distributed load 6 kN/m per meter together with concentrated loads of 30 kN at the left end A and 45 kN at the right end B. The supports are so located that the reaction is the same at each support. Determine the position of the props and draw B.M. and S.F. diagrams. Mark the values of the maximum B.M. and S.F.

Solution: Let the left support be at C and the right support be at D

Let AC = a meters

$$\therefore BD = 20 - 12 - a = (8 - a) \text{ meters}$$

Let R_1 and R_2 be the reactions at the left and right supports respectively.

$$\text{Total load on the beam} = 35 + 45 + (6 \times 20) \text{ kN} = 200 \text{ kN}$$

Since the reactions at the supports are given to be equal, we have

$$R_1 = R_2 = 100 \text{ kN}$$

Taking moments about A, we have ,

$$45 \times 20 + 6 \times 2 \times 10 = 100a + 100 (12 + a)$$

$$900 + 1200 = 100a + 1200 + 100a$$

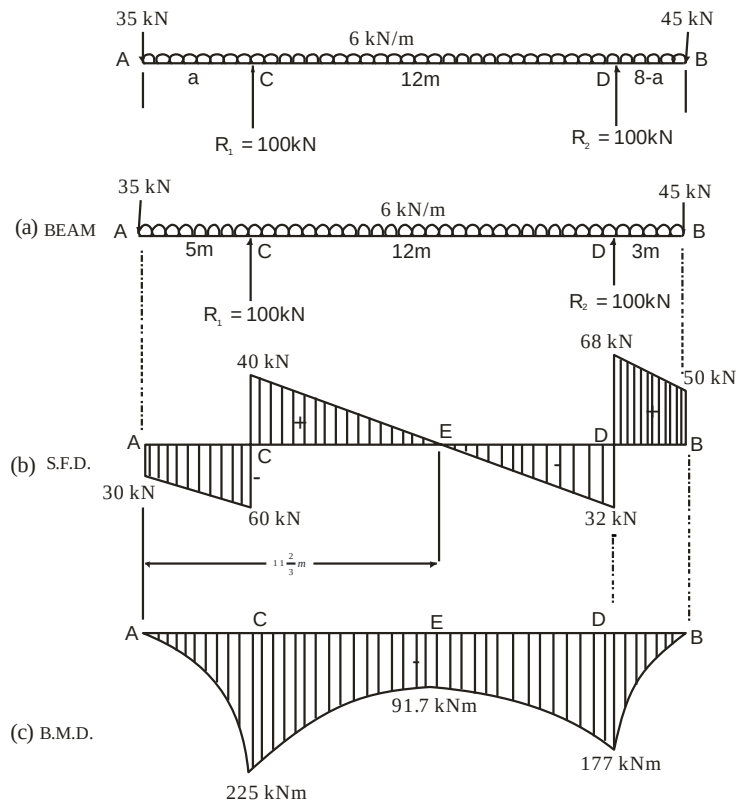


Fig. 3.25

$$200a = 900$$

$$\therefore a = 4.5 \text{ meters}$$

$\therefore$ The left support is at 4.5 meters from A and the right support is at $8 - 4.5 = 3.5$ meters from B.

Shear force calculations

S.F. just on the right hand side of A	=	- 30 kN
S.F. just on the left hand side of C	=	- 30 - 6 $\times$ 4.5 = - 57 kN
S.F. just on the right hand side of C	=	- 57 + 100 = + 43 kN
S.F. just on the right hand side of D	=	+ 45 + 6 $\times$ 3.5 = 66
S.F. just on the left hand side of D	=	+ 66 - 100
	=	- 34 kN

Let the S.F. be zero at a distance of x meters from A (between the two supports)

Equating the S.F. to zero, we have,

$$100 - 35 - 6x = 0$$

$$\therefore 6x = \frac{65}{6} = 10 \frac{5}{6}$$

$$\therefore x = \frac{65}{6} = 10 \frac{5}{6} \text{ m}$$

Bending moment calculations

$$\text{B.M at A} = 0$$

$$\text{B.M. at C} = M_d = -35 \times 4.5 - 6 \times \frac{4.5^2}{2} \text{ kNm} = -218.25 \text{ kNm}$$

$$\text{B.M. at D} = M_d = -45 \times 3.5 - 6 \times \frac{3.5^2}{2} \text{ kNm} = -194.25 \text{ kNm}$$

B.M. at distance of $10\frac{5}{6}$ meter from A

$$= M_e = 100 \times \frac{5}{6} - 30 \times \frac{10\frac{5}{6}}{6} - \frac{6 \left(10\frac{5}{6}\right)^2}{2} \text{ kNm} = -93.75 \text{ kNm}$$

Problem 3.12 Calculate the reactions for the beam shown in Fig. 3.26 Construct the bending moment and shear force diagrams. Determine the location of the maximum bending moment and mark it clearly on each of the diagrams.

Solution: Let R_a and R_c be the reactions at the supports A and C respectively.
Taking moments about A, we have

$$R_c \times 5 = 80 \times \frac{3}{2} + 15 \times 7$$

$$\therefore R_c = 45 \text{ kN}$$

$$\therefore R_a = 80 + 15 - 45 = 50 \text{ kN}$$

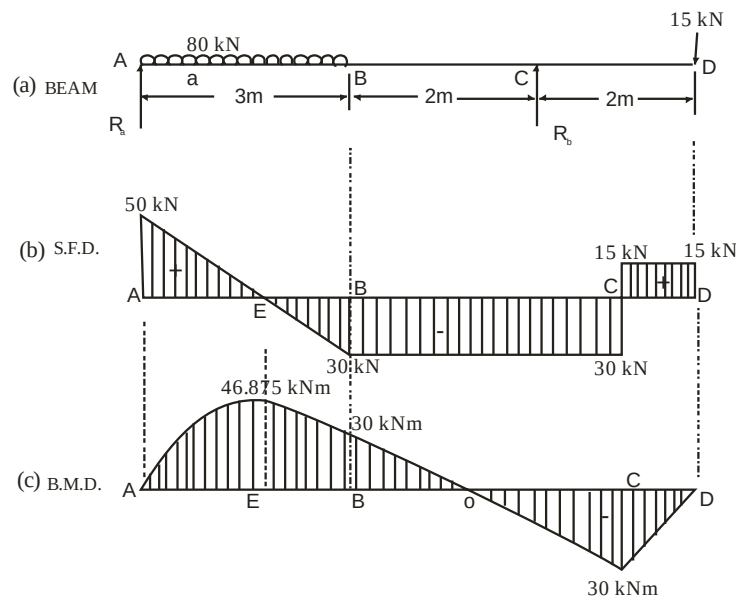


Fig. 3.26

S.F. Calculations

$$\text{S.F. at A} = +50 \text{ kN}$$

$$\text{S.F. at B} = 50 - 80 = -30 \text{ kN}$$

$$\text{S.F. between B and C} = -30 \text{ kN}$$

$$\text{S.F. between C and D} = +15 \text{ kN}$$

Let the S.F. be the zero at x meters from A.

Equating the S.F. to zero, we have,

$$50 - \frac{80}{3}x = 0$$

$$\therefore x = \frac{3}{80} \times 50 = 1.875 \text{ m from A}$$

B.M. Calculations

$$\text{B.M. at A} = M_a = 0$$

$$\text{B.M. at B} = M_b = 50 \times 3 - 80 \times \frac{3}{2} \text{ kNm} = + 30 \text{ kNm}$$

$$\text{B.M. at E i.e., 1.875 meters from A} = 50 \times 1.875 - \frac{80 (1.875)^2}{2} \text{ kNm} = + 46.875 \text{ kNm}$$

$$\text{B.M. at C} = M_c = - 15 \times 2 = - 30 \text{ kNm}$$

$$\text{B.M. at D} = M_d = 0$$

Point of contra flexure

There will be a point of contraflexure between B and C.

Let this point of contraflexure be at x meter from C

Equating the B.M. to zero, we get,

$$40x - 15 (2 + x) = 0$$

$$\therefore x = 1.2 \text{ m}$$

Exercise

3.1 What are the different types of beams?

3.2 What are the sign conventions for shear force and bending moment?

3.3 What do you mean by point of contraflexure?

3.4 What are the different types of load acting on a beam?

3.5 Draw SFD and BMD for a cantilever subjected to udl.

3.6 A cantilever of length 2 m carries a udl of 3 kN/m over a length of 1 m from the fixed end. Draw SFD and BMD.

(Ans. $F_{\max} = 3\text{kN}$, $M_{\max} = -1.5\text{kNm}$)

3.7 A cantilever of length 4 m carries a udl of 1 kN/m over whole length and a point load of 2 kN at 1 m from the free end. Draw SFD and BMD.

(Ans. $F_{\max} = 14\text{kN}$, $M_{\max} = -14\text{kNm}$)

3.8 A simply supported beam of length 8 m carries point loads of 4, 10 and 7 kN at a distance of 1.5, 2.5 and 2 m respectively from left end. Draw the SFD and BMD.

(Ans. $M_{\max} = 90\text{kNm}$)

3.9 A simply supported beam of length 8 m carries point loads of 4 and 6 kN at a distance of 2 and 4 m respectively from left end. Draw the SFD and BMD.

(Ans. $M_{\max} = 20\text{kNm}$)

3.10 A simply supported beam of length 6 m carries point loads of 1, 2 and 3 kN at a distance of 1.5, 3 and 4.5 m respectively from left end. There is a udl of 1.5 kN/m over the entire length. Draw the SFD and BMD.

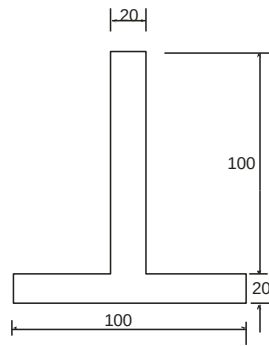
(Ans. $M_{\max} = 12.75\text{kNm}$)

Problem 4.1 Find the centroid of the section shown in Fig. 4.10(a).

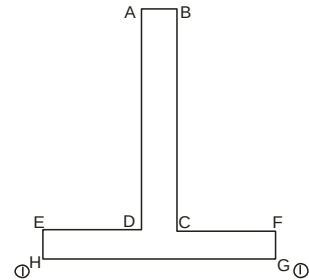
Solution: The section will be split up into two rectangular areas ABCD and EFGH as shown in Fig. 4.10(b) of areas,

$$20 \times 100 = 2000 \text{ mm}^2$$

and $100 \times 20 = 2000 \text{ mm}^2$, respectively



(a)



(b)

Fig. 4.10

Centroidal distance of ABCD from the axis 1 – 1 = 70 mm

Centroidal distance of EFGH from the axis 1 – 1 = 10 mm

Let $\bar{y}$ be the height of the centroid of the section from the axis 1 – 1

$$\bar{y} = \frac{\sum Ay}{\sum A} = \frac{2000 \times 70 + 2000 \times 10}{2000 + 2000} \text{ mm}$$

$$= 40 \text{ mm above the axis 1 – 1}$$

The above computation may be conveniently worked out in a tabular form as shown below:

Component	Area A mm ²	Centroidal Distance From 1 – 1 y mm	Ay mm ²
ABCD	2000	70	140000
EFGH	2000	10	20000
Total	4000		60000

$$\therefore \bar{y} = \frac{\sum Ay}{\sum A} = \frac{160000}{4000} = 40 \text{ mm}$$

Problem 4.2 Find the centroid of the section in Fig. 4.11.

Solution: The given section will be split up into a number of components. The area of the various components and their centroidal distance from axis 1 – 1 and the moments of the individual components about the axis 1- 1 are shown in the following table:

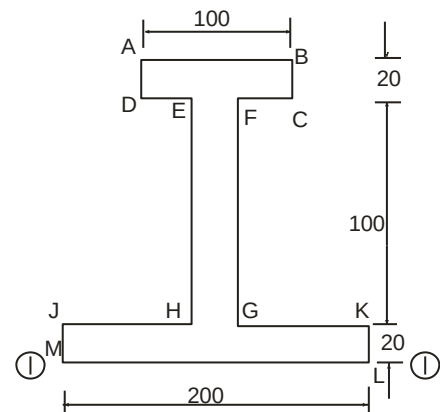


Fig. 4.11

Component	Area A mm ²	Centroidal Distance From y 1 – 1 mm	Ay mm ²
ABCD 100 × 20	2000	130	260000
EFGH 10 × 100	1000	70	70000
JKLM 200 × 20	4000	10	40000
Total	7000		370000

$$\bar{y} = \frac{\sum Ay}{\sum A} = \frac{370000}{7000} \text{ mm} = 52.86 \text{ mm above the axis 1 – 1.}$$

Problem 4.3 Find the centroid of the section shown in Fig. 4.12.

Solution: The given section may be split up into two rectangles ABCD and EFGH as shown in

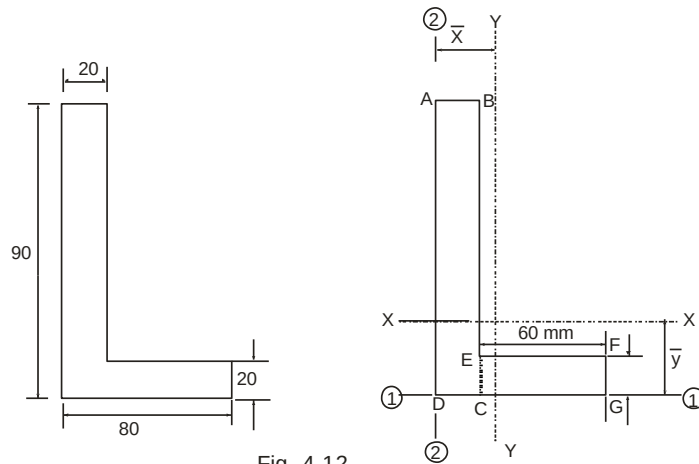


Fig. 4.12

Fig. 4.12. The position of the centroid of the section with respect to the axis 1 – 1 and 2 – 2 will now be worked out. The relevant computations are shown in the following table:

Component	Area A mm ²	Centroidal Distance y From 1 – 1 mm	Centroidal Distance x From 2 – 2 mm	Ay mm ³	Ax mm ³
ABCD 20 × 90	1800	45	10	81000	18000
EFGC 60 × 20	1200	10	50	12000	60000
Total	3000			93000	78000

$$\bar{y} = \frac{\sum Ay}{\sum A} = \frac{93000}{3000} = 31 \text{ mm}$$

$$\bar{x} = \frac{\sum Ax}{\sum A} = \frac{78000}{3000} = 26 \text{ mm}$$

Problem 4.4 Find the moment of inertia of the section shown in Fig. 4.13 about the centroidal axis perpendicular to the web.

Solution: The given section will be split up into components ABCD, EFGH and JKLM. The areas of the individual components, their centroidal distances from the axis 1 – 1 through the bottom edge and the moments of inertia of the individual components about their respective centroidal axes parallel to the axis 1 – 1 are given in the following table:

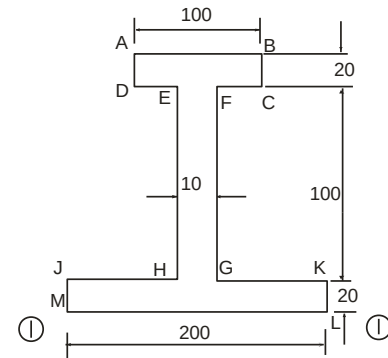


Fig. 4.13

Component	Area A mm ²	Centroidal distance y from 1 – 1 mm	Ay mm ³	Ay ² mm ⁴	I _{self} mm ⁴
ABCD	2000	130	260000	33800000	$\frac{100 \times 20^3}{12} = 66667$
EFGH	1000	70	70000	4900000	$\frac{10 \times 100^3}{12} = 833333.33$
JKLM	4000	10	40000	400000	$\frac{200 \times 20^3}{12} = 133333$
TOTAL	7000		370000	39100000	1033333.33

$$\text{Distance of the centroidal axis XX from the axis 1-1} \\ = \bar{y} = \frac{\sum Ay}{\sum A} = \frac{370000}{7000} \text{ mm} = 52.86 \text{ mm}$$

$$\text{Moment of inertia about the axis 1 – 1} \\ = I_{1-1} = \sum I_{self} + \sum Ay^2 \\ = 1033333.33 + 3910.0000 = 4013333.33 \text{ mm}^4$$

$$\text{But } I_{1-1} = I_{xx} + \sum A\bar{y}^2 \\ \therefore 4013333.33 = I_{xx} + 7000 \times 52.86^2 \\ \therefore I_{xx} = 20574076.13 \text{ mm}^4$$

Problem 4.5 Find the moment of inertia about the centroidal axes XX and YY of the section shown in Fig. 4.14.

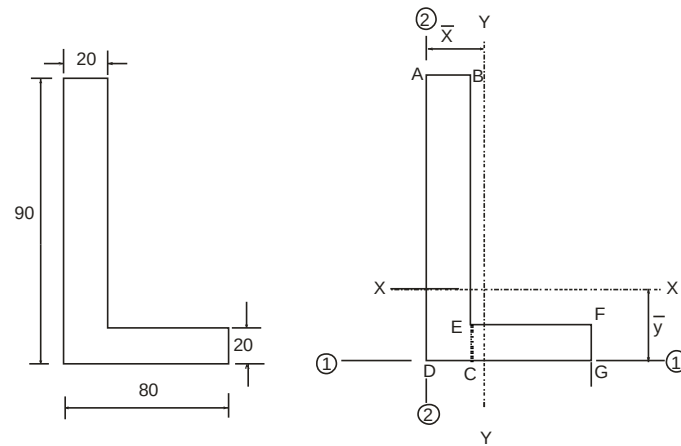


Fig. 4.14

Solution: To find the moment of inertia about the axis XX:

The following table shows the relevant computations:

Component	Area A mm ²	Centroidal distance y from 1 – 1 mm	Ay mm ³	Ay ² mm ⁴	I _{self} mm ⁴
ABCD	1800	45	81000	3645000	$\frac{20 \times 90^3}{12} = 1215000$
EFGH	1200	10	12000	120000	$\frac{60 \times 20^3}{12} = 40000$
TOTAL	3000		93000	3765000	1255000

$$\therefore \bar{y} = \frac{\sum Ay}{\sum A} = \frac{93000}{3000} = 31$$

$$\text{Moment of inertia about the axis 1 – 1} = I_{1-1} = \sum I_{self} + \sum Ay^2$$

$$\text{But } I_{1-1} = 1255000 + 3765000 = 5020000 \text{ mm}^4$$

$$5020000 = I_{xx} + 3000 \times 31^2$$

$$\therefore I_{xx} = 2137000 \text{ mm}^4$$

To find the moment of inertia about the axis YY:

The following table shows the relevant computations:

Component	Area A mm ²	Centroidal distance x from 2 – 2	Ax mm ³	Ax ² mm ⁴	I _{self} mm ⁴
-----------	------------------------------	---	-----------------------	------------------------------------	--------------------------------------

		mm			
ABCD	1800	10	18000	180000	$\frac{90 \times 20^3}{12} = 60000$
EFGH	1200	50	60000	3000000	$\frac{20 \times 60^3}{12} = 360000$
TOTAL	3000		78000	3180000	420000

$$\therefore \bar{x} = \frac{\sum Ax}{\sum A} = \frac{78000}{3000} = 26 \text{ mm}$$

Moment of inertia about the axis 2 – 2

$$= I_{2-2} = \sum I_{self} + \sum Ax^2$$

$$\therefore I_{2-2} = 420000 + 3180000 = 3600000$$

But $I_{2-2} = I_{yy} + (\sum A)(\bar{x}^2)$

$$3600000 = I_{yy} + 3000 \times 26^2$$

$$I_{yy} = 1572000 \text{ mm}^4$$

Problem 4.6 Calculate the moment of inertia about the horizontal and vertical gravity axes (I_{xx} and I_{yy}) of the section shown in Fig. 4.15.

Solution: The given section is split up into two components. The areas of the individual components, their centroidal distances from the axis 1 – 1 through the top edge and the Moments of Inertia of the individual components about their centroidal axes parallel to the axis 1 – 1 are tabulated below:

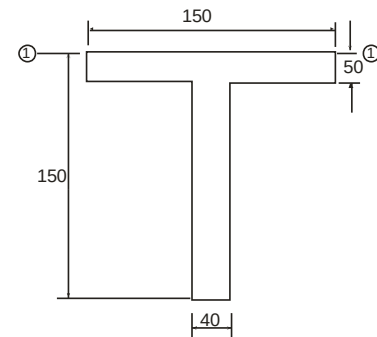


Fig. 4.15

Component	Area A mm ²	Centroidal distance y from 1 – 1 mm	Ay mm ³	Ay ² mm ⁴	I _{self} mm ⁴
Top Flange	7500	25	187500	4687500	$\frac{150 \times 50^3}{12} = 1562500$
Web	4000	100	400000	40000000	$\frac{40 \times 100^3}{12} = 3333333.3$
TOTAL	11500		587500	44687500	4895833.3

Distance of the centroidal axis XX from 1 – 1

$$= \bar{y} = \frac{\sum Ay}{\sum A} = \frac{587500}{11500} = 51.09 \text{ mm}$$

$$I_{1-1} = \sum I_{self} + \sum Ay^2 = 4895833.3 + 44687500 = 49583333.3$$

But $I_{1-1} = I_{xx} + (\sum A)\bar{y}^2$

$$49583333.3 = I_{xx} + 11500 \times 51.09^2$$

$\therefore$

$$I_{xx} = 19566170.15 \text{ mm}^4$$

$$= \frac{50 \times 150^3}{12} + \frac{100 \times 40^3}{12}$$

$$I_{yy} = \frac{50 \times 150^3}{12} + \frac{100 \times 40^3}{12} = 14595833.33 \text{ mm}^4$$

Problem 4.7 Find the moment of inertia about the centroidal axis XX for the section shown in

Fig. 4.16. The dimensions shown in the Fig. 4.16 are in cm.

Solution: The given section will be split up into three rectangular components, i.e., the top flange, the web and the bottom flange.

The properties of the components are tabulated below:

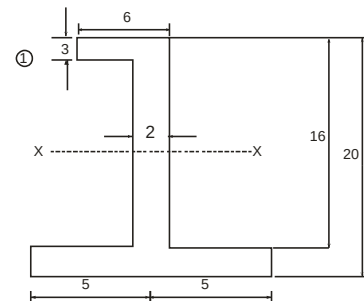


Fig. 4.16

Component	Area of (cm ²)	Centroidal distance y from 1 – 1 (cm) bottom edge	Ax (cm ³)	Ax ² (cm ⁴)	I _{self} (cm ⁴)
Top Flange	18	18.5	333	6160.50	$\frac{6 \times 3^3}{12} = 13.5$
Web	26	10.5	273	2866.50	$\frac{2 \times 16^3}{12} = 366.167$
Bottom flange	40	2	80	160	$\frac{10 \times 2^3}{12} = 53.333$
TOTAL	84		686	9187	433

Distance of centroidal axis XX from the bottom edge 1 – 1

$$= \bar{y} = \frac{\sum Ay}{\sum A} = \frac{686}{84} = 8.17 \text{ cm}$$

Moment of inertia about the axis 1

$$= I_{1-1} = \sum I_{self} + \sum Ay^2 = 433 + 9187 = 9620 \text{ cm}^4$$

But

$$I_{1-1} = I_{xx} + A \bar{y}^2$$

$$\therefore 9620 = I_{xx} + 84 \times 8.172$$

$$\therefore I_{xx} = 4015.38 \text{ cm}^4$$

Exercise

4.1 Define centre of gravity and centroid.

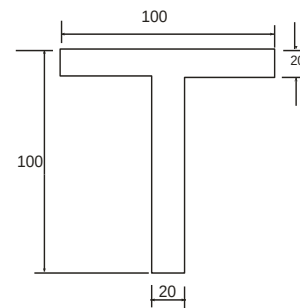
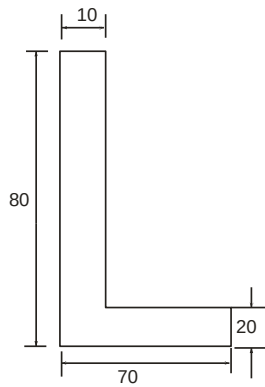
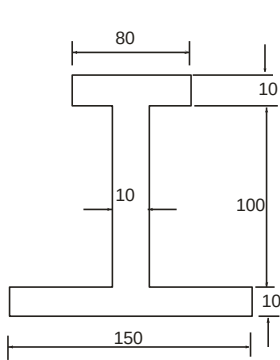
4.2 Define radius of gyration and moment of inertia.

4.3 Find the expression for moment of inertia of a rectangular section about the axis passing through its base.

4.4 State the parallel axis theorem and perpendicular axis theorem.

4.5 Find the expression for moment of inertia of a hollow rectangular section about the axis passing through its base.

4.6 Find the centre of gravity and moment of inertia of the following sections about their centroidal axis:



5. BENDING STRESSES IN BEAMS

5.1 Introduction

Due to external loads and moments bending moments and shearing forces are set up at all the sections of a beam. A beam section has to resist the action of bending moment and shearing force so induced and the beam deforms. The process of bending will stop when it has set up full resistance to the bending moment and shearing force.

The stress produced at the section to resist the bending moment is known as bending stress or direct stress or flexural stress and that to resist the action of shear force is known as shearing stress.

5.2 Theory of simple bending

Assumptions: Before developing elastic flexural formula it will be worthwhile to outline the assumptions upon which such relationship depends:

- (1) The material of the beam remains elastic during loading i.e. elastic limit is not exceeded.
- (2) The material of the beam is perfectly homogenous and isotropic and modulus of elasticity in tension is equal to that in compression.
- (3) Before loading takes place the beam is straight and each cross section is symmetric about plane of bending.
- (4) Plane cross section remains plane before and after bending.
- (5) There is no resultant force perpendicular to any cross section.
- (6) Every layer of material is free to expand or contract longitudinally and laterally under stress and does not exert pressure upon each other.

Pure Bending or Simple Bending:

Fig. 5.1 shows a beam AB with supports at C&D, L units apart. Let the overhangs AC = BD = a. Let a point load be applied at each end of the beam. It can be observed that there is a constant bending moment and no shear force (zero shear force) between C&D. The stresses will be set up in that length of the

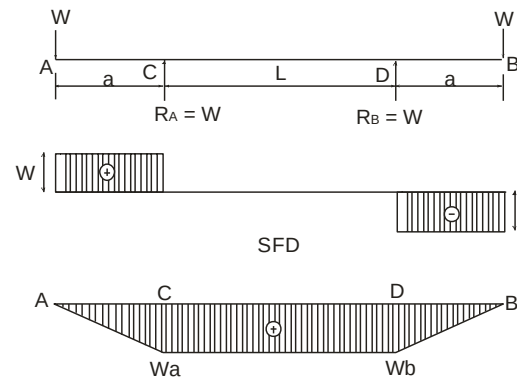


Fig. 5.1

beam due to bending moment only and that length of the beam is said to be in pure bending or simple bending.

Fig. 5.2 Shows the longitudinal section of a beam through plane of symmetry before and after it has been bent by two equal and apposite moments applied at its ends. The axes of coordinates are also shown in the same figure. The bent form of the longitudinal beam section shown in Fig. 5.2 (b) is highly exaggerated but it represents the essential features.

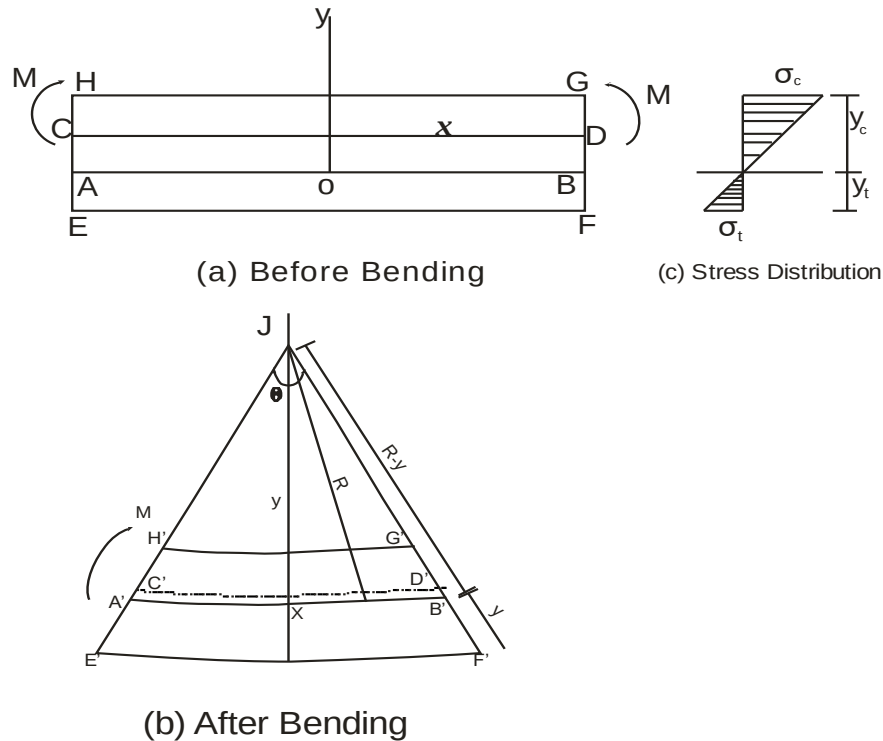


Fig. 5.2

Since the bending moment at any section is same, the section of the beam under consideration will bend uniformly or the curvature of any surface will be uniform as in Fig. 5.2 b. Thus the sides of longitudinal section GH and EF will bend into concentric arcs of circles, $G'H'$ and $E'F'$. All surfaces between and parallel to GH and EF will bend into contracted arcs of circles with centre at J . From Fig. 5.2(b) it can be seen that while $G'H'$ has contracted from its original length GH , the lower fiber $E'F'$ has extended from its original length EF . If we go down from $G'H'$ to $E'F'$, the contraction reduces becomes zero at some layer and then the layer begins to extend.

The layer of beam at which the extension or contraction does not take place, i.e. the layer which remains in its original length after bending is termed as or neutral layer. Line

AB in Fig. 5.2(a) and arc A'B' in fig. 5.2(b) represent neutral layer. The line of intersection of neutral layer and beam's cross section is defined as neutral axis.

The Fig. 5.2(b) shows the position of beam after bending. Let the radius of the neutral layer after and before bending be R so that the radius of layer C'D' is 'R-y'. It may be noted that AB = CD before bending and AB = A'B' after bending being neutral layer. C'D' has shortened from its original length CD. If the angle subtended by concentric arcs at common centre J is θ then

$$AB = CD = A'B' = R \theta$$

and $C'D' = (R - y) \theta$

strain of fiber CD $e = \frac{\text{Change in length}}{\text{Original Length}} = \frac{C'D' - A'B'}{A'B'}$

$$e = \frac{(R - y)\theta - R\theta}{R\theta} = \frac{-y}{R} \quad (\text{compressive}) \quad \dots(5.1)$$

The negative sign indicates that strain is compressive. If another layer at a distance y below the neutral plane is considered, e will be positive or tensile.

Equation $e = \frac{-y}{R}$ shows strain varies directly as the distance from neutral axis

because R is constant for a given loading.

Since stress = strain x elasticity

$$\sigma = e \times E = \frac{y}{R} E$$

$$\frac{\sigma}{y} = \frac{E}{R} \quad \dots\dots (5.2)$$

or $\sigma \propto y$

$\sigma_t \propto y_t$

$\sigma_c \propto y_c$

Hence the stress intensity in any fiber is proportional to the distance of the fiber from the neutral layer. For the case explained above all fibers below the neutral layer are subjected to tensile stresses while those above the neutral layer are subjected to compressive stresses. The stress distribution diagram is shown in Fig. 5.2(c).

Moment of Resistance:

Let us now determine the magnitude of stress from the equation of equilibrium, since there is no external force, the sum of all the compressive forces above the neutral layer must be equal to the sum of the tensile forces below the neutral layer.

Let us take a strip δy thick at a distance y from NA in Fig 5.3 and let its area be δa . Since the stress is equal to ' σ ' there the thrust exerted by this small area will be ' $\sigma \cdot \delta a$ '.

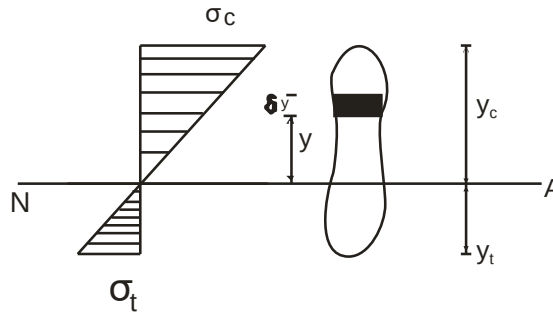


Fig. 5.3 Stress Distribution

To find M_r let us again consider the elementary area δa in Fig. 5.3. The force set up by the area $\sigma \delta a$ and the moment of force about NA is

$$\delta M = \sigma \cdot \delta a \cdot y = \frac{E}{R} (\delta a) \cdot y^2$$

The total moment of resistance set up by such elementary area will be

$$\sum_{y=y_t}^{y=y_c} \frac{E}{R} \delta a \cdot y^2 = \frac{E}{R} \int_{y_t}^{y_c} y^2 \delta a$$

The expression in the integral is known as second moment of area or moment of inertia about NA and is shown as 'I'

Hence $M_r = \frac{E}{R} I$

for equilibrium $M = M_r = \frac{E}{R} I$

$$\frac{M}{I} = \frac{E}{R} \quad \text{.....(5.4)}$$

But we know $\frac{E}{R} = \frac{\sigma}{y}$

$$\frac{M}{I} = \frac{E}{R} = \frac{\sigma}{y} \quad \text{.....(5.5)}$$

$$\frac{M}{I} = \frac{\sigma}{y}$$

For tensile side use σ_t and y_t

$$\frac{M}{I} = \frac{\sigma}{y}$$

For compressive side use σ_c and y_c

'EI' is termed as *flexural rigidity* and may be specified as a single constant for a given beam.

Eq. 5.5 is applicable to a member which is subjected to a constant bending moment and the member is absolutely free from shear force but in actual practice, a member is subjected to such loading that the B.M. varies from section to section and also the shear force is not zero. But shear force is zero at a section where bending moment is maximum. Hence the condition of simple bending may be assumed to be satisfied at such a section.

5.3 Design Criterion And Section Modulus

The bending equation is quite significant while designing machine components. To ensure strength in a beam it is necessary that the greatest tensile and compressive bending stress at the critical section i.e. where bending moment is maximum should not exceed the corresponding allowable stress for the material.

$$\sigma_{tmax} = \frac{M}{I} y_t ;$$

$$\sigma_{cmax} = \frac{M}{I} y_c$$

Condition of strength or of design

$$\sigma_{tmax} \leq \sigma_t \text{ allowable and } \sigma_{cmax} \leq \sigma_c \text{ allowable}$$

for a given value of allowable stress

$$\sigma = \frac{M}{I} y$$

$$M = \sigma \frac{I}{y} = \sigma Z \quad [\text{Compare with } F = \sigma.A \text{ for tensile loading}] \dots\dots(5.6)$$

Where $Z = \frac{I}{y}$ and is known as *section modulus*. Greater the value of Z stronger will be the section.

5.4 Section Modulus For Various Shapes Of Beam Section :

5.4.1 Rectangular Section

Moment of inertia of a rectangular section (Fig 5.4) about an axis through its C.G. (or through NA) is given by,

$$I = \frac{bd^3}{12}$$

Distance of outer most layers from N.A. is given by,

$$y_{\max} = \frac{d}{2}$$

∴ Section modulus is given by

$$Z = \frac{I}{y_{\max}} = \frac{\frac{bd^3}{12}}{\left(\frac{d}{2}\right)}$$

$$= \frac{bd^3}{12} \times \frac{2}{d} = \frac{bd^2}{6}$$

.....(5.7)

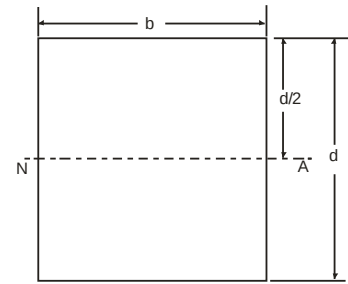


Fig. 5.4

5.4.2 Hollow Rectangular Section

Moment of inertia of a hollow rectangular section (Fig 5.5) about an axis through its C.G. (or through NA) is given by,

Here $I = \frac{BD^3}{12} - \frac{bd^3}{12}$

$$= \frac{1}{12} [BD^3 - bd^3]$$

$$y_{\max} = \frac{D}{2}$$

$$\therefore Z = \frac{I}{y_{\max}} = \frac{\frac{1}{12} [BD^3 - bd^3]}{\left(\frac{D}{2}\right)}$$

$$= \frac{1}{6D} [BD^3 - bd^3]$$

.....(5.8)

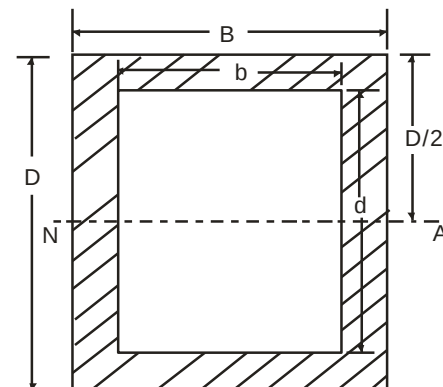


Fig. 5.5

5.4.3 Circular Section

Moment of inertia of a circular section (Fig 5.6) about an axis through its C.G. (or through NA) is given by,

$$I = \frac{\pi}{64} d^4$$

and

$$y_{\max} = \frac{d}{2}$$

$$\frac{I}{y_{\max}} = \frac{\frac{\pi}{64} d^4}{\left(\frac{d}{2}\right)}$$

∴

$$Z =$$

$$= \frac{\pi}{32} d^3$$

.....(5.9)

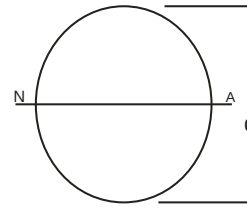


Fig. 5.6

5.4.4 Hollow Circular Section

Moment of inertia of a hollow circular section (Fig 5.7) about an axis through its C.G. (or through NA) is by,

Here $I = \frac{\pi}{64} [D^4 - d^4]$

and

$$y_{\max} = \frac{D}{2}$$

$$\begin{aligned} \therefore Z &= \frac{I}{y_{\max}} = \frac{\frac{\pi}{64} [D^4 - d^4]}{\left(\frac{D}{2}\right)} \\ &= \end{aligned}$$

$$= \frac{\pi}{32D} [D^4 - d^4]$$

.....(5.10)

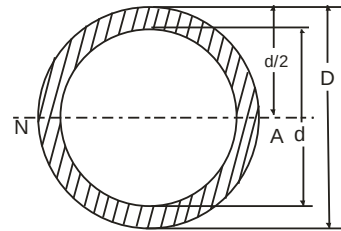


Fig. 5.7

5.7)
given

Problem 5.1 Copper plate is bent into a circular arc of radius 10 meters. If the plate section be 120 mm wide and 20 mm thick, find the maximum stress induced and the bending moment which can produce this stress. Take $E = 1 \times 10^5 \text{ N/mm}^2$.

Solution: Moment of inertia of the section about the neutral axis.

$$I = \frac{12 \times 20^3}{12} = 80000 \text{ mm}^4$$

$$\frac{M}{I} = \frac{\sigma}{y} = \frac{E}{R}$$

$$\sigma = \frac{E}{R} \cdot y$$

$$\therefore \sigma_{\max} = \frac{1 \times 10^5}{10 \times 10^3} \times \left(\frac{20}{2} \right) \text{ N/mm}^2 = 100 \text{ N/mm}^2$$

$$M = \frac{E}{R} \cdot I = \frac{1 \times 10^5}{10 \times 10^3} \times 8 \times 10^4 = 8 \times 10^5 \text{ N mm}$$

$$= 800 \text{ Nm}$$

Problem 5.2 A timber joist of 200 mm width and 400 mm depth is used as a cantilever beam of 3 meters length and loaded with a concentrated load at its free end in addition to its self-weight, so that the bending stress does not exceed 7 N/mm^2 . Find out the maximum value of the applied load. Unit weight of timber is 5 kN/m^3 .

Solution: Let W (N) be the concentrated load that can be placed safely at the free end of the cantilever under the given conditions of the problem. W = self weight of the joint per meter run.

$$= 0.2 \times 0.4 \times 1 \times 5 = 0.4 \text{ kN/m} = 400 \text{ N/m}$$

I_{xx} = Moment of inertia of the section about N.A.

$$= \frac{12 \times 200 \times 400^3}{12} = 32 \times 10^8 \text{ mm}^4$$

M = Maximum bending moment which will occur at the fixed end.

$$= WL + \frac{WL^2}{2}$$

$$= W \times 3 + \frac{100 \times 3 \times 3}{2} \text{ Nm} = (3W + 450) \times \frac{10^3}{400} \text{ N mm}$$

$$y = \text{Distance of extreme fiber from N.A.} = \frac{400}{2} = 200 \text{ mm.}$$

$$\sigma = \text{Bending stress} = 7 \text{ N/mm}^2$$

Using the relation $\frac{M}{I} = \frac{\sigma}{y}$ and substituting the values

$$\frac{(3W + 450) \times 10^3 \times 3}{32 \times 10^8} = \frac{7}{200}$$

or

$$3W + 450 = 37333.33$$

$\therefore$

$$W = 12294.44 \text{ N}$$

Problem 5.3 A cast iron T-beam has the following dimensions:

Overall depth	= 160 mm
Width of flange	= 150 mm
Flange thickness	= 40 mm
Web thickness	= 50 mm

The beam is simply supported over a span of 2.5 meters placed in the inverted T-position (i.e. flange at the bottom). If the maximum allowable tensile stress in the flange and compressive stress in the web are limited to 25 N/mm² and 80 N/mm² respectively, find the maximum central load that the beam can safely carry.

Solution: As the Fig. 5.8 is symmetrical about Y-Y axis, the centre of gravity will lie on this axis. Let $\bar{y}$ be the distance of centre of gravity from the bottom of flange. Divide the figure into two rectangles as shown in Fig. 5.8.

Rectangle 1

$$a_1 = 150 \times 40 = 6000 \text{ mm}^2, y_1 = \frac{40}{2} = 20$$

mm

Rectangle 2

$$a_2 = 120 \times 50 = 6000 \text{ mm}^2, y_2 = 40 + \frac{120}{2}$$

= 100 mm

$$\bar{y} = \frac{a_1 y_1 + a_2 y_2}{a_1 + a_2} = \frac{6000 \times 20 + 6000 \times 100}{6000 + 6000}$$

= 60 mm

Moment of inertia of the section about NA

$$= \frac{1}{12} \times 150 \times 40^3 + 150 \times 40 \times (60 - 20)^2 + \frac{1}{12} \times 50 \times 120^3 + 50 \times 120 \times (100 - 60)^2$$

$$= 272 \times 10^5 \text{ mm}^4$$

σ_t = Permissible tensile stress at the bottom most face of flange = 25 N/mm²,
at a distance of $y_t = 60$ mm from neutral axis.

σ_c = Permissible compressive stress at the top most face of web = 80 N/mm²,
at a distance of $y_c = 100$ mm from neutral axis.

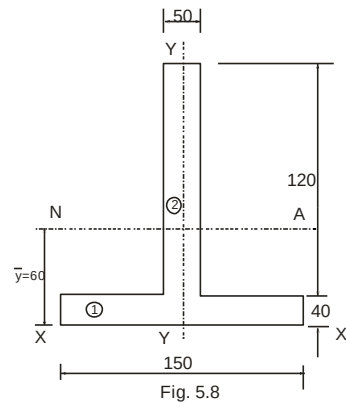
$$\frac{\sigma_t}{y_t} = \frac{25}{60} = 0.42, \quad \frac{\sigma_c}{y_c} = \frac{80}{100} = 0.80$$

$\frac{\sigma_t}{y_t}$

y_t being the least, is to be taken into consideration. Let W N be the concentrated load

that can be safely carried under the given condition of the problem. Maximum bending

moment at the centre, $M = \frac{Wl}{4} = \frac{4 \times 2.5}{4} \text{ Nm}$.



$$\frac{M}{I} = \frac{\sigma_t}{y_t} \text{ and substituting the values}$$

$$\frac{W \times 2.5 \times 1000}{4 \times 272 \times 10^5} = 0.42$$

$$\therefore w = 18278.4 \text{ N}$$

Problem 5.4 A 4 meters long beam with rectangular section of 100 mm width and 200 mm depth is simply supported at the ends. If it is loaded with a uniformly distributed load of 4 kN/m throughout the span and a concentrated load $P = 2 \text{ kN}$ placed at a distance of 1.5 m from one end, determine the maximum bending stress in the beam.

Solution: Let R_A and R_B be the reactions at A and B respectively. Taking moment of all the forces about A:

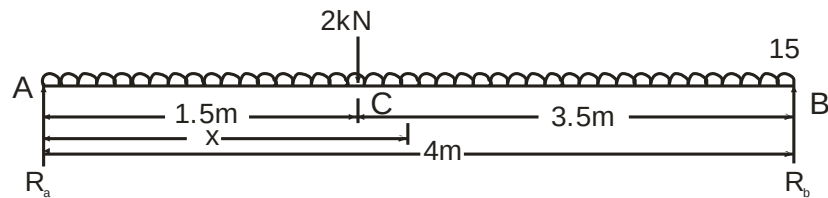


Fig. 5.9

$$R_B \times 4 - 4 \times 4 \times \frac{4}{2} - 2 \times 1.5 = 0 \quad \therefore R_B = 8.75 \text{ kN.}$$

$$\text{But } R_A + R_B = 4 \times 4 + 2 = 18$$

$$\therefore R_A = 18 - 8.75 = 9.25 \text{ kN.}$$

Let us consider a section at a distance x from A.

$$M_x = 9.25x - 2(x - 1.5) - 4 \cdot x \cdot \frac{x}{2} = 7.25x + 3 - 2x^2$$

M_x is maximum when

$$\frac{d}{dx} (7.25x + 3 - 2x^2) = 0$$

$$\text{or } 7.25 - 4x = 0 \quad \therefore x = 1.81 \text{ m}$$

$$\text{Moment bending of the section, } M = \frac{7.25 \times 1.81 + 3 - 2 \times (1.81)^2}{100 \times (200)} = 9.5703 \text{ kN.m}$$

$$\text{Moment of inertia of the section, } I = \frac{12}{100 \times (200)^3} = 6666.666 \times 10^4 \text{ mm}^4$$

$$\therefore \text{Maximum bending stress, } \sigma = \frac{M}{I} \times y = \frac{9.5703 \times 10^3 \times 10^3}{6666.66 \times 10^4} \times 100$$

$$= 14.35 \text{ N/mm}^2$$

Problem 5.5 A floor has to carry a load of 10000 N/sq.m. It is supported by joists 150 mm $\times$ 300 mm over a span of 5 m. How far apart may the joists be placed so that the bending stress does not exceed 8 N/mm²?

Solution: Let x be the centre-to-centre spacing of the joists. The area of the floor between any two joists = $5 \times x \text{ m}^2$.

The total load supported by one interior joist = $5 \times x \times 10000 = 50000x \text{ N}$.

$$\text{Maximum bending moment, } M = \frac{wl^2}{8} = \frac{50000x \times 5^2}{8} \text{ Nm.}$$

$$\text{Moment of inertia of the joist, } I = \frac{bd^3}{12} = \frac{0.15 \times (0.30)^3}{12} = 3.375 \times 10^{-4} \text{ m}^4$$

$$\frac{I}{y} \times \sigma = \frac{M}{y}$$

$$\text{We know, } M = \frac{50000x \times 5^2}{8} = 18000 \text{ Nm}$$

Equating this to the maximum bending moment

$$\frac{50000x \times 5^2}{8} = 18000$$

$$\therefore x = 0.576 \text{ m}$$

Problem 5.6 Compare the weight of two equally strong beams of circular sections made of the same material, one being of solid section and the other of hollow section with inside diameter $\frac{2}{5}$ of the external diameter.

Solution: For solid let the diameter be d

For hollow section: Let d_1 = Outside diameter

d_2 = Inside diameter

$$\therefore \frac{d_1}{d_2} = \frac{5}{2}$$

$$\text{For solid section, } Z_s = \frac{\pi d^3}{32}$$

$$\frac{\pi(d_1^4 - d_2^4)}{32d_1} = \frac{\pi d_1^3}{32} \left[1 - \left(\frac{2}{5} \right)^4 \right] = \frac{609\pi d_1^3}{32 \times 625}$$

$$\text{For hollow section, } Z_H =$$

$$\text{As both the beams are equally strong, } Z_s = Z_H$$

$$\frac{\pi d^3}{32} = \frac{609\pi d_1^3}{32 \times 625} \quad \text{or} \quad \frac{d_1^3}{d^3} = \frac{625}{609}$$

$$\therefore \frac{d_1}{d} = 1.00868$$

Both the beams are made of same material. Therefore $\rho_s = \rho_H = \rho$

$$\text{Let } W_s = \text{weight of solid shaft} = \text{volume} \times \text{density}$$

$$= \frac{\pi}{4} d^2 \times L \times \rho$$

$$W_H = \text{Weight of hollow shaft} = \text{volume} \times \text{density}$$

$$= \frac{\pi}{4} \left[d_1^2 - \left(\frac{2}{5} d_1 \right)^2 \right] \times L \times \rho$$

$$\therefore \frac{W_s}{W_H} = \frac{\frac{\pi}{4} d^2 \times L \times \rho}{\frac{\pi}{4} \left[d_1^2 - \left(\frac{2}{5} d_1 \right)^2 \right] \times L \times \rho} = \frac{d^2}{d_1^2 - \frac{4}{25} d_1^2}$$

$$\frac{W_s}{W_H} = \frac{25d^2}{21d_1^2} = \frac{25}{21} \times \frac{1}{(1.00868)^2} = 1.17$$

Therefore the weight of solid shaft is 1.17 times the weight of the hollow shaft.

Problem: 5.7 Fig. 5.10 shows an I-section proposed to be used for a beam of span 8 meters. The beam carries a uniformly distributed load of 100 kN/m over the whole span. Find the extreme bending stress for the section and state if the stress is permissible. If not, find the width of a 15 mm thick cover plate for each flange for the section to be safe. Find also the length of this cover plate. Safe stress in bending is 160 N/mm².

Solution: Moment of inertia of the I-section³

$$= I = \frac{250 \times 640^3}{12} - \frac{235 \times 600^3}{12} \text{ mm}^4$$

$$= 1.2313 \times 10^9 \text{ mm}^4$$

Maximum bending moment for the beam

$$= M = \frac{wl^2}{8} = \frac{100 \times 8^2}{8} = 800 \text{ kNm}$$

Extreme stress due to bending moment

$$= f = \frac{M}{I} y = \frac{800 \times 10^6}{1.2313 \times 10^9} \times 320 \text{ N/mm}^2$$

$$= 208 \text{ N/mm}^2$$

This is greater than the permissible stress of 160 N/mm². Suppose a cover plate of 15 mm thickness of provided for each flange, Increase in moment of inertia

$$= 2 \left[\frac{b \times 15^3}{12} + b \times 15 \times 327.5^2 \right]$$

$$= 3218250 b \text{ mm}^4$$

Moment of inertia of the modified section

$$= I' = 1.2313 \times 10^9 + 3218250 b \text{ mm}^4$$

Limiting the extreme stress to the permissible value of 160 N/mm²

$$f = 160 = \frac{M}{I'} y'$$

$$\frac{800 \times 10^6 \times 327.5}{I'}$$

$$\therefore I' = \frac{160}{160} = 1.63752 \times 10^9 \text{ mm}^4$$

$$\therefore 1.2313 \times 10^9 + 3218250 b = 1.63752 \times 10^9$$

$$\therefore b = 126.2 \text{ mm say } 130 \text{ mm}$$

Moment of resistance of the I section alone

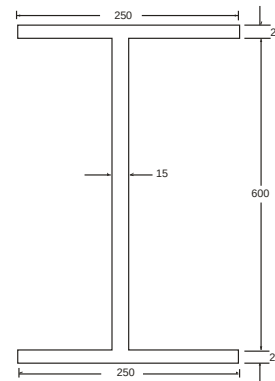


Fig. 5.10

$$= \frac{I}{y} \times f = \frac{1.2313 \times 10^9}{320} \times 160 = 6.1565 \times 10^8 \text{ Nmm} = 615.65 \text{ kNm}$$

Let the bending moment for the girder reach 615.65 kNm at a distance of x meter from the support

$$\therefore \frac{100 \times 10}{2} x - 100 \times \frac{x^2}{2} = 615.65$$

$$\therefore 50x^2 - 500x + 615.65 = 0$$

$$\therefore x^2 - 10x + 12.31 = 0 \quad \therefore 1.44 \text{ m from each end.}$$

Hence length of cover plate required

$$\therefore = 10 - 1 \times 1.44 = 7.124 \text{ m}$$

Problem 5.8 A timber beam is freely supported on supports 6 meters apart. It carried a uniformly distributed load of 12 kN per meter run and a concentrated load of 9 kN at 2.5 meters from the left support. If the stress in timber is not to exceed 10 N/mm², design a suitable section making the depth twice the width.

Solution: Fig. 5.11 shows the beam carrying the loading mentioned in the problem. Let R_a and R_b be the reactions at the left and right supports.

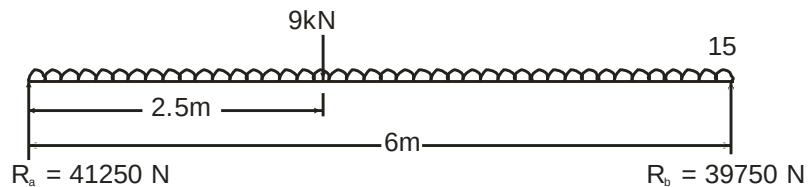


Fig. 5.11

Taking moments about the left support A, we have

$$R_b \times 6 = 12000 \times 6 \times 3 + 9000 \times 2.5$$

$$\therefore R_b = 39750 \text{ N}$$

$$\therefore R_a = 12000 \times 6 + 9000 - 39750 \text{ N} = 41250 \text{ N}$$

Let the shear force be zero at x meter from B

$$\therefore 39750 - 12000x = 0$$

$$\therefore x = \frac{39750}{12000} = 3.3125 \text{ m}$$

$\therefore$ Maximum bending moment occurs at 3.3125 meters from B

$$\therefore \text{Maximum bending moment} = 39750 \times 3.3125 - \frac{12000 \times (3.3125)^2}{2} \text{ Nm}$$

$$= 65836 \text{ Nm} = 65836 \times 1000 \text{ Nmm}$$

But the moment of resistance of the section

$$= \frac{1}{6} \sigma_b b d^2$$

Since $\sigma_b = 10 \text{ N/mm}^2$ and $b = \frac{d}{2}$, we have,

$$\frac{1}{6} \times 10 \times \frac{d}{2} \times d^2 = \frac{65836 \times 1000}{10}$$

$$d^3 = \frac{65836 \times 1000 \times 12}{10}$$

$$\begin{aligned} \therefore d &= 369.64 \text{ mm} \cong 370 \text{ mm} \\ \therefore b &= \frac{d}{2} = \frac{370}{2} = 184.8 \text{ mm} \cong 185 \text{ mm} \\ \therefore \text{The beam section is } 231 \text{ mm} \times 462 \text{ mm} \end{aligned}$$

Problem 5.9 Find the width x of the flange of a cast iron beam having the section shown in Fig. 5.12 so that the maximum compressive stress is three times the maximum tensile stress the member being in pure bending subjected to sagging moment. The depth h of the beam is 80 mm, the thickness t of the web and flange is 20 mm.

Solution: Let the distance of the centroidal neutral axis from the bottom edge be $\bar{y}$. Let σ_c and σ_t be the maximum compressive and maximum tensile stress

$$\therefore \frac{\sigma_c}{\sigma_t} = \frac{\bar{y}}{h - \bar{y}} = 3$$

$$\therefore \frac{h - \bar{y}}{\bar{y}} = 3$$

$$\therefore \bar{y} = \frac{h}{4}$$

But from the geometry of the section,

$$\bar{y} = \frac{th \cdot \frac{h}{2} + (x - t)t \cdot \frac{t}{2}}{th + (x - t)t} = \frac{h}{4}$$

$$\therefore x = t + \frac{h^2}{h - 2t}$$

Putting, $t = 20 \text{ mm}$, and $h = 80 \text{ mm}$

$$\therefore x = 20 + \frac{80^2}{80 - 2 \times 20} \text{ mm}$$

$$\therefore x = 180 \text{ mm}$$

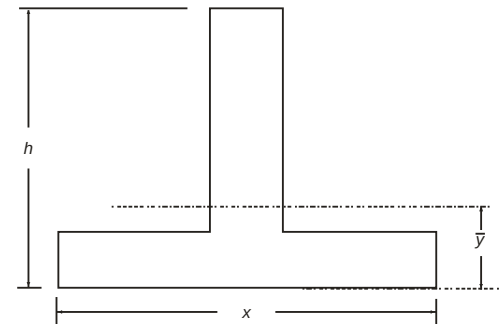


Fig. 5.12

Problem 5.10 A cast iron bracket subject to bending has cross-section of I-form with unequal flanges. The depth of the section is 280 mm and the metal is 40 mm thick throughout. The top flange is 200 mm wide and the bottom flange is 100 mm wide. Find the position of the neutral axis and the moment of inertia of the section about the neutral axis and determine the maximum bending moment that should be imposed on this section if the tensile stress in the top flange is not to exceed 20 N/mm^2 . What is then the value of the maximum compressive stress in the bottom flange?

Solution: Fig. 5.13 shows the section of the bracket. The section may be conveniently divided into three rectangular components. The areas of the individual components, their centroidal distance from the top edge and their moments about the top edge and their moments of inertia about their centroidal axes are tabulated below:

Component	Area a mm^2	Centroidal distance y from top edge y (mm)	ay (mm) ³	ay^2 (mm) ⁴	I_{self} (mm) ⁴
Top Flange	8000	20	16×10^4	320×10^4	$\frac{200 \times 40^3}{12} = 106.67 \times 10^4$
Web	8000	140	112×10^4	15680×10^4	$\frac{40 \times 200^3}{12} = 2666.67 \times 10^4$
Bottom Flange	4000	260	104×10^4	27040×10^4	$\frac{100 \times 40^3}{12} = 53.33 \times 10^4$
TOTAL	20000		232×10^4	43040×10^4	2826.67×10^4

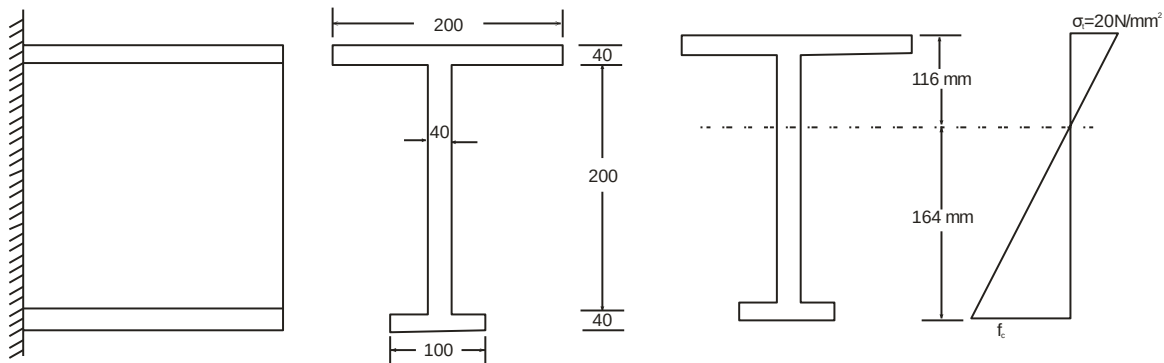


Fig. 5.13

$$\therefore \text{Distance of the neutral axis from the upper edge}$$

$$= \bar{y} = \frac{\sum ay}{\sum a} = \frac{232 \times 10^4}{20000} = 116 \text{ mm}$$

Moment of inertia about the upper edge

$$= I_{uu} = \sum I_{self} + \sum ay^2$$

$$2826.67 \times 10^4 + 43040 \times 10^4 = 45866.67 \times 10^4 \text{ mm}^4$$

But

$$I_{uu} = I_{xx} + (\sum a)\bar{y}^2$$

$$45866.67 \times 10^4 = I_{xx} + 20000 \times 116^2$$

$$\therefore I_{xx} = (45866.67 - 26912) \times 10^4 = 18954.67 \times 10^4$$

Let the maximum bending moment be in Nmm.

$$\frac{M}{I} = \frac{\sigma}{y_t}$$

$$\frac{\sigma_t}{y_t} \times I$$

$\therefore$

$$M = \frac{\sigma_t}{y_t} \times I \quad (\text{But } \sigma_t = 20 \text{ N/mm}^2)$$

$$= \frac{20}{116} \times 18954.67 \times 10^4 \text{ Nmm}$$

$$= 3268 \times 10^4 \text{ Nmm} = 32680 \text{ Nm}$$

Let the maximum compressive stress be $\sigma_c = \frac{164}{116} \times 20 \text{ N/mm}^2$

$$\therefore \sigma_c = 28.28 \text{ N/mm}^2$$

$$\therefore \sigma_c = 28.28 \text{ N/mm}^2$$

Problem 5.11 The following three beams have the same length and are of the same material and have same weight.

- (i) I-section 400 mm × 180mm with 20 mm thick flanges and 12.5 mm thick web.
- (ii) Rectangular section of depth twice its width.
- (iii) Solid circular section.

Compare the strength of the I-section with the rectangular and circular sections.

Solution: Obviously all the three beams have the same sectional sections area.

Properties of the I-section

$$\text{Area} = 2(180 \times 20) + (360 \times 12.5) = 11700 \text{ mm}^2$$

$$= 3.0876 \times 10^8 \text{ mm}^3$$

$$\text{Section modulus } Z_1 = \frac{3.0876 \times 10^8}{200}$$

$$= 1543800 \text{ mm}^3$$

Properties of the rectangular section

$$\text{Area} = 11700 \text{ mm}^2, \text{ Width} = b, \text{ depth} = d = 2b$$

$$\therefore \text{Area} = 2b^2 = 11700$$

$$\therefore b = 76.485 \text{ mm}$$

$$\text{and } d = 2 \times 76.485 = 152.97 \text{ mm}$$

$$\text{section modulus } Z_2 = \frac{bd^3}{6} = \frac{76.485 \times 152.97^3}{6} = 298289.22 \text{ mm}^3$$

Properties of circular section

$$\text{Area} = 11700 \text{ mm}^2, \text{ Diameter} = d$$

$$\therefore \text{Area} = \frac{\pi d^2}{4} = 11700 \quad \therefore d = 122.053 \text{ mm}$$

$$\text{Section modulus } Z_3 = \frac{\pi d^3}{32} = \frac{\pi \times 122.053^3}{32} = 178502.9 \text{ mm}^3$$

$$\text{Comparison of strength} = \frac{\text{Strength of I-section}}{\text{Strength of rectangular section}} = \frac{Z_1}{Z_2} = \frac{1543800}{298289.22} = 5.176$$

$$\frac{\text{Strength of I-section}}{\text{Strength of circular section}} = \frac{Z_1}{Z_3} = \frac{1543800}{178502.9} = 8.649$$

Exercise

5.1 Derive the expression for bending stress for a layer in a beam.

5.2 What do you mean by section modulus? Find an expression for section modulus for rectangular, circular and hollow circular section.

5.3 What is the meaning of 'strength of a section'?

5.4 What do you mean by 'simple bending' or 'pure bending'? What are the assumptions made in theory of simple bending?

5.5 A timber beam is 120 mm wide and 200 mm deep and is used on a span of 4m. the beam carries a udl of 2.8 kN/m run over the entire length. Find the maximum bending stress induced.

(Ans. 7 N/mm²)

5.6 A rectangular beam 10 m long is simply supported at its ends and is acted upon by a 10 kN/m udl throughout its length. Find the size of the beam if the thickness is twice that of its width. Take the allowable bending stress as 40 N/mm².

(Ans. b = 167.3 mm, d = 334.6 mm)

5.7 A rectangular beam 300mm deep is simply supported over a span of 4m. what udl per meter the beam may carry if the bending stress is not to exceed 120 N/mm². Take $I = 8 \times 10^6 \text{ mm}^4$.

(Ans. 3.2 kN/m)

5.8 For a given stress compare the moments of resistance of a beam of square section when placed (a) with its two sides horizontal (ii) with its diagonal horizontal.

(Ans. 1.414)

5.9 A 10m long C.I. pipe of 500mm internal diameter and 20 mm thickness is simply supported at its ends. Find the maximum intensity of bending stress induced in the metal when the pipe is running full with water. Take unit weights of C.I. and water as 70 kN/mm³ and 10 N/mm³.

(Ans. 13.17 N/mm²)

5.10 A beam of I-section 400mm deep has flanges 200 mm wide and 30 mm thick and web 24 mm thick. Compare the flexural strength with that of a beam rectangular section of the same weight, depth being twice the breadth.

(Ans. 3.27)